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SUPPLEMENT TO "HAL-MLE LOG-SPLINES DENSITY ESTIMATION (PART I: UNIVARIATE)"

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S.1. Notation, Definition, and Representation.

S.1.1. *Notation.*

S.1.1.0.1. Acronym.
TV: Total Variation
BTV: Bounded Total Variation
SVN: Sectional Variational Norm
BSVN: Bounded Sectional Variational Norm
HAL: Highly Adaptive Lasso
TMLE: Target Maximum Likelihood Estimator (or Minimum Loss)
HAL-TMLE: TMLE that uses HAL for initial estimation
HAL-MLE: Highly Adaptive Lasso Maximum Likelihood Estimator
KDE: Kernel Density Estimation
LAS: Local Adaptive Splines
TF: Trend Filtering
TFPP: Trend Filtering with Penalty on the Parametric Part
LSDE: Logspline Density Estimation
PLSDE: TV-penalized Logspline Density Estimation
KM: Kaplan-Meier Estimator
CV: Cross Validation
càdlàg function: multivariate real-valued function on (say) unit cube $[0,1]^d$ that is right-continuous with left-hand limits
CDF: generalized cumulative distribution function.
DGP: Data-Generating Process.
LLFP: local least favorable path
ULFP: universal least favorable path
FISTA: Fast Iterative Shrinkage-Thresholding Algorithm
L-BFGS: Limited-memory Broyden-Fletcher-Goldfarb-Shanno
AdaGrad: Adaptive Gradient

S.1.1.0.2. Mathematical Notation.
$\sim$: We say $X \sim P_0$ if the random variable X has distribution P_0.
$\lesssim$: We say $a \lesssim b$ if $a \leq Cb$ for some constant C.
$\asymp$: We say $a \asymp b$ if $cb \leq a \leq Cb$ for some constant c and C.
$\asymp^+$: We say $a \asymp^+ b$ if $cb \leq a \leq Cb(\log n)^p$ for some constant c, C, and finite p. Note that p can be negative.
$\equiv$: Definition.
$\mathrm{TV}(\cdot)$: Total variation functional as defined in Section 9.1 of Tibshirani et al. (2022). When

the function f is a càdlàg function, $\mathrm{TV}(f)=\int_0^1 |df(u)|$. This is also referred to as the total variation norm (a seminorm), denoted as $|f|_v$.
X: Euclidean valued observed random variable.
P_0: True data-generating distribution of X or the truth expectation operator in empirical process.
P_n: Empirical data-generating distribution of X or the empirical expectation operator in empirical process.
P_{f_n}: Expectation operator with measure Q_n.
P_{f_0}: Expectation operator with measure f_0, which is the truth P_0.
$(\mathbb{G}_n(f): f \in \mathcal{F})$: denotes empirical process $\mathbb{G}_n = n^{1/2}(P_n - P) \in \ell^\infty(\mathcal{F})$ indexed by a class of functions $\mathcal{F}$.
$\mathcal{M}$: Statistical model, the set of possible probability distributions for P_0.
$O^+(r(n))$: We use notation $O^+(r(n))$ if the term is $O(r(n)(\log n)^p)$ for some finite p, and the $+$ notation is also used in O_p^+. In words, it is described as "up till a $\log n$ factor".
$d_0(f,f_0)$: loss-based dissimilarity between f and f_0 defined by $d_0(f,f_0)=P_0L(f)-P_0L(f_0)$ for some general loss L.
$P_0 \in \mathcal{M}$: P_0 is known to be an element of the statistical model $\mathcal{M}$.
$\mathrm{d}f/\mathrm{d}\mu$: Radon-Nikodym derivative of measure implied by cadlag function f w.r.t. Lebesgue measure μ.
$p=\mathrm{d}P/\mathrm{d}\mu$: Density of P w.r.t. dominating measure μ
p_0: True density of data-generating distribution P_0 w.r.t. appropriate dominating measure μ
p_f: The density p indexed by a function f through a link function.
$\|f\|_v$: variation norm of cadlag function f.
$f^{(k)}$: k-th order Lebesgue-Radon-Nikodym derivative of cadlag function f.
$N(\varepsilon,\mathcal{F},\|\cdot\|)$: the covering number of class $\mathcal{F}$ w.r.t. some metric defined as the minimal number of balls with radius ε needed to cover $\mathcal{F}$.
$J(\delta,\mathcal{F},\|\cdot\|)$: the entropy integral for class $\mathcal{F}$ w.r.t. some metric defined as $\int_0^\delta \sqrt{\log N(\varepsilon,\mathcal{F},\|\cdot\|)}\mathrm{d}\varepsilon$. $J_\infty(\delta,\mathcal{F})$ is used when the norm is the supremum norm. $J_2(\delta,\mathcal{F})$ is used when the norm is the L^2 norm.
Π: projection operator.
S_f: score operator at f.
$D^*(P)$: canonical gradient (efficient influence curve) of a pathwise differentiable parameter at P.

S.1.1.0.3. Definition.
f_n: we use subscript n for estimations, like β_n or C_n. Here in this paper, f_n means the HAL-MLE.
$f_{n,\beta_n(M)}$: the HAL-MLE with a user-specified M. So it is parameterized by the coefficient β_n under the L_1 constraint less than M. $f_{n,\beta_n(M)}$ is the MLE over $D_M^{(k)}(\mathcal{R}_n)$
M_{cv}: the hyperparameter M chosen by CV.
f_n^{rel}: Relaxed HAL-MLE. Relax the HAL-MLE $f_{n,\beta_n(M)}$ by loosening the L_1 constraint with a larger constant $C>M$, such that it is a MLE over $D_C^{(k)}(\mathcal{R}_n)$. Notice that we use the word "fully relaxed" when we drop the L_1 constraint completely
Undersmooth: The behavior that we intentionally include more basis to reduce the bias, which will increase variance. Practically, we can choose $M' > M_{cv}$ to realize this. It represents the trade off between bias and variance.
$f_{n,0}$: The oracle HAL-MLE over a sample of size n. Such that $f_{n,0}=\arg\min_{f\in D_M^{(k)}(\mathcal{R}_{n,0})} -P_0 \log p_f$.
$\tilde{f}_n$: The projection of f_n to some other linear subspace. Typically, we used $\tilde{f}_n$ to denote the

projection of f_n on $D^{(k)}_M(\mathcal{R}_{n,0})$ in Appendix S.4.
f_n^k: the k-th step TMLE update, shorthand notation of f_{n,ε_n^k}. f_n is also f_n^0.
$f^{(k)}$: the k-th order derivative of f.
C_n: The normalizing constant of $\tilde{f}_n$, also denoted as $C(\tilde{f}_n)$, or $C(\tilde{\beta}_n)$.
$C_{n,0}$: The oracle normalizing constant, also denoted as $C(f_{n,0})$, or $C(\beta_{n,0})$.
$\|f\|_v^*$: sectional variation norm $\|f\|_v^* = \int_{[0,1]^d} |df(u)|$.
$D^{(0)}_U([0,1]^d)$: Space of real valued cadlag functions on $[0,1]^d$ with BSVN for some $U < \infty$. We use $D^{(0)}_U([0,1])$ for our univariate case.
$\|f\|_{v,k}^*$: k-th order sectional variation norm of f.
$D^{(k)}_U([0,1])$: functions f in $D^{(0)}_U([0,1])$ for which all k-th order Lebesgue-Radon-Nikodym derivatives exist and $\|f\|_{v,k}^* < U < \infty$
$\mathcal{R}_0$: complete infinite index set of basis.
$\mathcal{R}_n$: a finite data-adaptive index set of basis. The active set after Lasso selection. For the nonparametric part of the basis system, we index them each by a tuple (k,u) for k-th order splines and any knot point $u \in (0,1]$. For the parametric part of the basis system, we index them each by a tuple $(i,0)$, for $i \le k$ and starting point 0. As such, we could define the total index set as a union of all these.
$D^{(k)}(\mathcal{R}_n)$: for a given subset $\mathcal{R}_n \subset \mathcal{R}_0$, this is defined as the space of functions that are in closure of linear span of $\{\phi_{k,u} : (k,u) \in \mathcal{R}_n\}$.
$D^{(k)}_M(\mathcal{R}_n)$: the subset of $D^{(k)}(\mathcal{R}_n)$ with a user-specified M as the L_1 norm constraint of the basis.
$D^{(k)}_M(\mathcal{R}_{n,0})$: an independent oracle space with a user-specified M as the L_1 norm constraint of the basis. A technique used for proving pointwise asymptotic normality.
J_n: the effective dimension of $D^{(k)}_M(\mathcal{R}_n)$. The number of active basis.
$J_{n,0}$: the effective dimension of $D^{(k)}_M(\mathcal{R}_{n,0})$. The number of active basis.
ϕ_j^*: some orthonormal basis function on $D^{(k)}_M(\mathcal{R}_{n,0})$.
$\bar{\phi}_{n,x}$: a special basis function at point x defined as $\bar{\phi}_{n,x} = J_{n,0}^{-1/2} \sum_{j\in\mathcal{R}_{n,0}} \phi_j^*(x)\phi_j^*$. Used interchangably with $\bar{\phi}$.
$\tilde{\phi}_j$: the projection of ϕ_j^* onto some linear span, like $D^{(k)}_M(\mathcal{R}_n)$. In general, we use the tilde sign above an element to denote its projection on some other spaces, e.g. $\tilde{\phi}_{n,x}$, $\tilde{f}_n$.

S.1.2. *Representation Theorem.* This is the proof for Proposition 2.3 in the main paper.

PROOF. *Base case:* For $k=0$ (with $u_0 = u$),

$$f(x) = f(0) + \int_{(0,x]} \mathrm{d}\, f(u) = f(0) + \int_{(0,1]} I\{u \le x\}\, \mathrm{d}\, f(u) = f(0) + \int_0^1 (x-u)_+^0\, \mathrm{d}\, f(u).$$

Induction: If, for a k-times differentiable function f, the representation holds true, we now show it for a $(k+1)$-times differentiable f.

Since $f^{(k)}$ is differentiable,

$$\mathrm{d}\, f^{(k)}(u_k) = f^{(k+1)}(u_k)\, \mathrm{d}u_k, \quad f^{(k+1)}(u_k) \in D^{(0)}_U([0,1]).$$

Starting from the inductive hypothesis,

$$f(x) = \sum_{i=0}^{k} \frac{1}{i!} f^{(i)}(0)\, x^i + \int_0^1 \frac{1}{k!}(x-u_k)_+^k\, \mathrm{d}\, f^{(k)}(u_k)$$

$$
\begin{aligned}
&= \sum_{i=0}^{k} \frac{1}{i!} f^{(i)}(0)\, x^i + \int_0^1 \frac{1}{k!}(x-u_k)_+^k\, f^{(k+1)}(u_k)\,\mathrm{d}u_k \\
&= \sum_{i=0}^{k} \frac{1}{i!} f^{(i)}(0)\, x^i + \int_0^1 \frac{1}{k!}(x-u_k)_+^k \\
&\quad \times \Big(f^{(k+1)}(0) + \int_0^{u_k} \mathrm{d}\, f^{(k+1)}(u_{k+1}) \Big)\,\mathrm{d}u_k \\
&= \sum_{i=0}^{k} \frac{1}{i!} f^{(i)}(0)\, x^i + \int_0^1 \frac{1}{k!}(x-u_k)_+^k\, f^{(k+1)}(0)\,\mathrm{d}u_k \\
&\quad + \int_0^1 \frac{1}{k!} \int_0^{u_k} (x-u_k)_+^k\, \mathrm{d}\, f^{(k+1)}(u_{k+1})\,\mathrm{d}u_k \\
&= \sum_{i=0}^{k} \frac{1}{i!} f^{(i)}(0)\, x^i + f^{(k+1)}(0) \int_0^x \frac{1}{k!}(x-u_k)^k\,\mathrm{d}u_k \\
&\quad + \int_0^1 \frac{1}{k!} \int_0^{u_k} (x-u_k)_+^k\, \mathrm{d}\, f^{(k+1)}(u_{k+1})\,\mathrm{d}u_k \\
&= \sum_{i=0}^{k+1} \frac{1}{i!} f^{(i)}(0)\, x^i + \int_0^1 \frac{1}{k!} \int_0^{u_k} (x-u_k)_+^k\, \mathrm{d}\, f^{(k+1)}(u_{k+1})\,\mathrm{d}u_k \\
&= \sum_{i=0}^{k+1} \frac{1}{i!} f^{(i)}(0)\, x^i \\
&\quad + \int_0^1 \frac{1}{k!} \int_0^1 (x-u_k)^k I\{u_{k+1} \le u_k \le x\}\, \mathrm{d}\, f^{(k+1)}(u_{k+1})\,\mathrm{d}u_k \\
&= \sum_{i=0}^{k+1} \frac{1}{i!} f^{(i)}(0)\, x^i \\
&\quad + \int_0^1 \frac{1}{k!} \int_0^1 (x-u_k)^k I\{u_{k+1} \le u_k \le x\}\,\mathrm{d}u_k\, \mathrm{d}\, f^{(k+1)}(u_{k+1}) \\
&\quad \text{(by Fubini's theorem)} \\
&= \sum_{i=0}^{k+1} \frac{1}{i!} f^{(i)}(0)\, x^i + \int_0^1 \frac{1}{k!} \Big(\int_{u_{k+1}}^x (x-u_k)^k\,\mathrm{d}u_k \Big)\, \mathrm{d}\, f^{(k+1)}(u_{k+1}) \\
&= \sum_{i=0}^{k+1} \frac{1}{i!} f^{(i)}(0)\, x^i + \int_0^1 \frac{1}{(k+1)!}(x-u_{k+1})_+^{k+1}\, \mathrm{d}\, f^{(k+1)}(u_{k+1}).
\end{aligned}
$$

Then, by induction, the representation holds for all k. □

S.2. L^2 Convergence Analysis. We start with the L^2 convergence proof, which is a standard proof based on the convergence of loss-based dissimilarity. Let $\mathcal{F} := D_U^{(k)}([0,1])$, and define $f_0 = \arg\min_{f\in\mathcal{F}} P_0 L(f)$. Let M be the user-supplied L_1 norm constraint and $\mathcal{R}_n$ be the index set of the finite-dimensional working model, and define $f_n = \arg\min_{f\in D_M^{(k)}(\mathcal{R}_n)} P_n L(f)$. Define the loss-based dissimilarity to be $d_0(f_n, f_0) =$

$P_0L(f_n) - P_0L(f_0)$. Notice that $d_0(f_n, f_0)$ is just the KL divergence of the true density p_{f_0} and the estimated density p_{f_n} when L is $-\log p_f$. Knowing the fact that KL divergence is equivalent to the squared L^2-norm when the positivity assumption in Theorem 4.6 of the main paper holds true, we can show the L^2 convergence of HAL-MLE by showing the convergence of $d_0(f_n, f_0)$. Again, the positivity assumption is guaranteed by the link function.

S.2.1. *Proof of Theorem 4.3 in the main paper.* We show the convergence by discussing the covering number of $D_U^{(k)}([0,1]^d)$ and its relationship to the covering number of $\mathcal{F}_L := \{L(f) - L(f_0) : f \in \mathcal{F}\}$. The sup–norm covering number of $D_M^k[0,1]^d$ is covered in Lemma 29 of (van der Laan, 2023), and this implies the L^2 covering number of $D_M^k[0,1]^d$. However, for the univariate case, it is sufficient—and conventional—to rely directly on existing L^2 results for functions whose k^{th} derivative has bounded total variation. The equivalency is already stated in Section 2.2 of the main paper. The earliest source we have been able to locate is (Birman and Solomjak, 1967), which establishes the sharp bound $\log N\Big(\varepsilon,\ D_U^{(k)}([0,1]),\ L^2(\mu)\Big) \lesssim \left(\frac{C}{\varepsilon}\right)^{\frac{1}{k+1}}$, where C is a constant. Similar results and follow-up applications also appear in (Mammen and Van De Geer, 1997), (Tibshirani, 2014), and (Sadhanala and Tibshirani, 2019). Consequently, we adopt the one-dimensional L_2 entropy bound throughout; it is fully compatible with—and indeed implied by the broader sup–norm result in (van der Laan, 2023).

To show the connection between $N\Big(\varepsilon,\ \mathcal{F},\ L^2(\mu)\Big)$ and $N\Big(\varepsilon,\ \mathcal{F}_L,\ L^2(P_0)\Big)$, we list two key assumptions.

ASSUMPTION S.2.1 (Quadratic loss-based dissimilarity assumption). For the loss function L and loss–based dissimilarity $d_0(f, f_0) = P_0L(f) - P_0L(f_0)$, assume

$$\sup_{f\in\mathcal{F}} \frac{P_0\big\{L(f) - L(f_0)\big\}^2}{d_0(f, f_0)} = O(1) < \infty.$$

REMARK S.2.2. When $L(f) = -\log p_f$ (negative log-likelihood), then Assumption S.2.1 becomes a fact van der Laan, Dudoit and Keles (2004).

ASSUMPTION S.2.3 (Norm comparison). Let $\|\cdot\|$ be the norm used for $N\Big(\varepsilon,\ \mathcal{F},\ \|\cdot\|\Big)$. Assume

$$\sup_{f\in\mathcal{F}} \frac{P_0\big\{L(f) - L(f_0)\big\}^2}{\|f - f_0\|^2} = O(1) < \infty.$$

REMARK S.2.4. If $\|\cdot\|$ is chosen as either $L_2(\mu)$ or L_∞, then Assumption S.2.1 implies Assumption S.2.3, based on the equivalency of KL divergence and the squared L^2 distance under positivity assumption.

THEOREM S.2.5 (Covering number equivalency). *For a choice of $\|\cdot\|$ such that Assumption S.2.3 holds (e.g. L^2 and L_∞), Assumption S.2.3 implies that $N\Big(\varepsilon,\ \mathcal{F},\ \|\cdot\|\Big) \asymp N\Big(\varepsilon,\ \mathcal{F}_L,\ L^2(P_0)\Big)$.*

PROOF. Let $\{f_j^\varepsilon : j = 1, \cdots, N(\varepsilon)\}$ be an ε-covering number on $\mathcal{F}$, then there $\exists j$, such that for any $f \in \mathcal{F}$, $\|f_j^\varepsilon - f\| < \varepsilon$. Notice that for negative log-likelihood loss, there is a one-to-one relationship between $\mathcal{F}$ and $\mathcal{F}_L$.

And this implies that any element in $\mathcal{F}_L$ can be expressed as $L(f) - L(f_0)$. Constructing a net $\{L(f_j^\varepsilon) - L(f_0) : j = 1, \cdots, N(\varepsilon)\}$ on $\mathcal{F}_L$ based on that covering number, then $\|((L(f_j^\varepsilon) - L(f_0)) - (L(f) - L(f_0))\|^2_{L^2(P_0)} = \|L(f_j^\varepsilon) - L(f)\|^2_{L^2(P_0)} = P_0\{L(f_j^\varepsilon) - L(f)\}^2 = O(\|f_j^\varepsilon - f\|^2) = O(\varepsilon^2)$. The second last step is implied by Assumption S.2.3. So this net is a covering number on $\mathcal{F}_L$ which indicates the equivalence of the covering number of both classes. □

Now we have developed all the tools needed for proving Theorem 4.3 in the main paper.

PROOF. Let $J(\delta, \mathcal{F}_L, L^2(P_0))$ be the entropy integral of $\mathcal{F}_L$; for brevity we write $J_2(\delta)$. Based on Theorem S.2.5, $\mathcal{F}_L$ and $\mathcal{F}$ shares the same entropy integral $J_2(\delta) = \delta^{\frac{2k+1}{2k+2}}$.

Define $\mathbb{G}_n = n^{1/2}(P_n - P_0)$ to be the empirical process. P_n is the empirical measure and P_0 the true data-generating measure. Formal definitions of covering numbers and entropy integrals can be found in Section 2.2 of Van Der Vaart et al. (1996).

$$
\begin{aligned}
0 \le d_0(f_n, f_0) &= -\big(P_n - P_0\big)\big\{L(f_n) - L(f_0)\big\} + P_n\big\{L(f_n) - L(f_0)\big\} \\
&\le -\big(P_n - P_0\big)\big\{L(f_n) - L(f_0)\big\}, \text{ because } f_n \text{ is the empirical risk minimizer} \\
&\le n^{-1/2} \sup_{\substack{f\in\mathcal{F}_L \\ \|f\|_{L^2(P_0)}\le\delta}} \big|\mathbb{G}_n(f)\big| \\
&\lesssim n^{-1/2} J_2(\delta)\left(1 + \frac{J_2(\delta)}{\delta^2 n^{1/2}}\right) \text{ Section 3.4.2 of (Van Der Vaart et al., 1996).}
\end{aligned}
$$

$J_2(\delta)$ dominates $\frac{J_2(\delta)^2}{\delta^2 n^{1/2}}$ when $J_2(\delta) \lesssim \delta^2 n^{1/2}$. For any $\delta \ge \delta_n \asymp n^{-\frac{k+1}{2k+3}}$, we can use $J_2(\delta)$ as the upper bounded for $\mathbb{E}_{P_0} \sup_{\substack{f\in\mathcal{F}_L \\ \|f\|_{L_2(P_0)}<\delta}} \big|\mathbb{G}_n(f)\big|$.

The sketch of the following proof is to create a sequence of $\delta \ge \delta_n$, denote as δ_k, to iteratively solve for the optimal rate. We start with the trivially large radius $\delta_0 = 1$ (and the corresponding $J(\delta_0) = O_p(1)$), which yields $d_0(f_n, f_0) = O_p\big(n^{-1/2}\big)$. By Assumption S.2.1,

$$
P_0\big\{L(f_n) - L(f_0)\big\}^2 = O_p\big(d_0(f_n, f_0)\big) = O_p\big(n^{-1/2}\big), \qquad \big\|L(f_n) - L(f_0)\big\|_{L_2(P_0)} = O_p\big(n^{-1/4}\big).
$$

Hence we update the radius to $\delta_1 \asymp n^{-1/4}$, which is still larger than or equal to δ_n. We repeat the argument, iteratively shrinking δ until convergence. At convergence the $(k+1)$-st update equals the k-th, i.e. $\delta_{k+1} = \delta_k = \delta$, and the fixed point solves

$$
\delta = \sqrt{n^{-1/2} J_2(\delta)} = n^{-1/4}\,\delta^{\frac{2k+1}{4k+4}} = \delta_n.
$$

Plugging δ_n back into $J_2(\delta)$ gives

$$
d_0(f_n, f_0) \;=\; O_p\big(n^{-1/2} J(\delta_n)\big) \;=\; O_p\big(n^{-\frac{2k+2}{2k+3}}\big).
$$

The positivity of density holds, which implies that $d_0(f_n, f_0) \asymp \|p_{f_n} - p_{f_0}\|^2_{L^2(\mu)}$; therefore

$$
\|p_{f_n} - p_{f_0}\|_{L^2(\mu)} = O_P\big(n^{-\frac{k+1}{2k+3}}\big).
$$

□

S.3. Score Equation Analysis for HAL-MLE. HAL-MLE, unlike the log-splines, is not a full parametric MLE on the working model. Instead, we have a L_1 norm constraint on the coefficients. So we pay a price that the empirical score equation is not exactly 0. In this appendix, we will show that the scores we need can be approximately solved with a negligible error.

Recall the definition of the score. Let $S_\beta(\phi_j) = \frac{\mathrm{d}}{\mathrm{d}\beta(j)} \log p_\beta$ be the score at β for $\beta(j)$, the j-th coefficient of β. We also denote this score as $S_{f_\beta}(\phi_j)$. Note that this equals

$$S_\beta(\phi_j) = \phi_j(X) - P_\beta \phi_j,$$

where $P_\beta\phi_j \equiv \int \phi_j(x) p_\beta(x)\mathrm{d}x$. We note $\phi \to S_\beta(\phi)$ is linear. HAL-MLE has one L_1 norm constraint and solves $J_n - 1$ scores. We name the linear span of these $J_n - 1$ scores as the L_1 constrained score space, and notice that this space is a closed linear subspace of $D^k(\mathcal{R}_n)$. In the latter proof in Appendix S.4 and Appendix S.5, we will construct $\bar{\phi}_{n,x}$ as defined before Theorem 4.4 of the main paper and we want to elaborate on the rate of $P_n S_{f_n}(\bar{\phi}_{n,x})$.

Let $\tilde{\phi}_{n,x}$ be the projection of $\bar{\phi}_{n,x}$ onto the L_1 constrained score space, that is, the linear span of scores $S_{f_n}(\tilde{\phi}_j)$, $j \in \mathcal{R}_n$ that HAL solves exactly so that $P_n S_{f_n}(\tilde{\phi}_j) = 0$ for $j \in \mathcal{R}_n$. Then, $P_n S_{f_n}(\tilde{\phi}_{n,x}) = 0$, and thus we have

$$P_n S_{f_n}(\bar{\phi}_{n,x}) = P_n\{S_{f_n}(\bar{\phi}_{n,x}) - S_{f_n}(\tilde{\phi}_{n,x})\}.$$

We note that the $L^2(P_{f_{n,0}})$-norm of $\bar{\phi}_{n,x}$ is bounded:

$$\|\bar{\phi}_{n,x}\|_{\beta_{n,0}} = J_{n,0}^{-1} \sum_{j\in\mathcal{R}_{n,0}} \{\phi_j^*(x)\}^2 = O(1). \tag{S.3.1}$$

We have $p_{f_{n,0}} > \delta > 0$ on $[0,1]$, so that we also have

$$\|\bar{\phi}_{n,x}\|_\mu = O(1). \tag{S.3.2}$$

Recall the L^2 Approximation Error Assumption between $D^k(\mathcal{R}_n)$ and $D^k_U([0,1])$ in Theorem 4.4 of the main paper

$$\sup_{f\in D_U^{(k)}([0,1])} \inf_{g\in D^{(k)}(\mathcal{R}_n)} \|f-g\|_\mu = O^+\big(J_n^{-(k+1)}\big).$$

The idea of this assumption is that we can select a rich enough set of basis functions that can approximate the truth in $L^2(\mu)$-norm. The L_1 constrained score space has only one less basis than $D^k(\mathcal{R}_n)$, so it is reasonable to extend this assumption to the L_1 constrained score space.

THEOREM S.3.1 (Negligibility of Score Approximation Error for HAL-MLE). *Suppose that the $L^2(\mu)$-Approximation Error Assumption can be extended to the L_1 constrained score space, then the empirical score equation satisfies*

$$n^{1/2} P_n S_{f_n}(\bar{\phi}_{n,x}) = o_P(1).$$

PROOF. If the $L^2(\mu)$-Approximation Error Assumption holds for $\bar{\phi}_{n,x}$ and its projection $\tilde{\phi}_{n,x}$ as well, then,

$$\|\bar{\phi}_{n,x} - \tilde{\phi}_{n,x}\|_{L^2(\mu)} = O^+(J_n^{-(k+1)}),$$

$$\begin{aligned} P_n S_{f_n}(\bar{\phi}_{n,x}) &= P_n\{S_{f_n}(\bar{\phi}_{n,x}) - S_{f_n}(\tilde{\phi}_{n,x})\} \\ &= (P_n - P_0)\{\bar{\phi}_{n,x} - \tilde{\phi}_{n,x}\} - (P_{f_n} - P_0)\{\bar{\phi}_{n,x} - \tilde{\phi}_{n,x}\} \end{aligned}$$

The first term can be bounded by an empirical process, where we shorthanded $\|\cdot\|_{L^2(\mu)}$ by $\|\cdot\|_\mu$,

$$
\begin{aligned}
(P_n - P_0)\{\bar{\phi}_{n,x} - \tilde{\phi}_{n,x}\} &= J_{n,0}^{\frac{1}{2}} n^{-\frac{1}{2}} \sup_{\substack{f \in \mathcal{D}_U^{(k)}([0,1]) \\ \|f\|_{L^2(P_0)} \leq \delta}} \left|\mathbb{G}_n(f)\right| \\
&= J_{n,0}^{\frac{1}{2}} n^{-\frac{1}{2}} J(\delta_n) \\
&= J_{n,0}^{\frac{1}{2}} n^{-\frac{1}{2}} \delta_n^{\frac{2k+1}{2k+2}} \\
&= O_P^+(J_{n,0}^{\frac{1}{2}} n^{-\frac{1}{2}} (J_n^{-1/k+1})^{(2k+1)/(2k+2)}) \\
&= O_P^+(n^{-\frac{1}{4k+6}} n^{-\frac{1}{2}} (n^{-\frac{2k+1}{4k+6}})) \\
&= O_P^+(n^{-\frac{4k+5}{4k+6}}) \\
&= o_p(n^{-\frac{1}{2}})
\end{aligned}
$$

We here used the same empirical process as in Appendix S.2, and the δ_n is the is the $L^2(P_0)$ convergencerate of $J_n^{-\frac{1}{2}}(\bar{\phi}_{n,x} - \tilde{\phi}_{n,x})$ going to 0. This rate is implied by the L^2 Approximation Error Assumption and the positivity of the true density p_0.

The second term can be bounded by Cauchy-Schwarz inequality.

$$
\begin{aligned}
(P_{f_n} - P_0)\{\bar{\phi}_{n,x} - \tilde{\phi}_{n,x}\} &\leq \|p_{f_n} - p_0\|_\mu \cdot \left\|\bar{\phi}_{n,x} - \tilde{\phi}_{n,x}\right\|_\mu \\
&= \|p_{f_n} - p_0\|_\mu \cdot \left\|\bar{\phi}_{n,x} - \tilde{\phi}_{n,x}\right\|_\mu \\
&= O_P^+(n^{-\frac{k+1}{2k+3}} J_n^{-(k+1)}) \\
&= O_P^+(n^{-\frac{2k+2}{2k+3}}) \\
&= o_p(n^{-\frac{1}{2}})
\end{aligned}
$$

And thus, we have

$$P_n S_{f_n}(\bar{\phi}_{n,x}) = O_P^+(n^{-\frac{2k+2}{2k+3}}),$$

and

$$n^{1/2} P_n S_{f_n}(\bar{\phi}_{n,x}) = o_P(1).$$

□

S.4. Proof of Pointwise Asymptotic Normality. This appendix includes the proof of Theorem 4.4, Theorem 4.6, and Corollary 4.7 in the main paper.

PROOF. Again, recall that the HAL-MLE over $D_M^{(k)}(\mathcal{R}_n)$ by

$$f_n \;=\; \arg \min_{f \in D_M^{(k)}(\mathcal{R}_n)} -P_n \log p_f = f_{\beta_n}.$$

For the proof, we need to build an auxiliary tool with a set $\mathcal{R}_{n,0}$ that is independent of P_n instead of $\mathcal{R}_n$. Let $\mathcal{R}_{n,0} = \mathcal{R}(P_n^{\#})$, using the same selection algorithm on an independent sample $P_n^{\#} \sim P_0$. Define the oracle MLE over $D_M^{(k)}(\mathcal{R}_{n,0})$ as

$$f_{n,0} = \arg \min_{f \in D_M^{(k)}(\mathcal{R}_{n,0})} -P_0 \log p_f \; = \; f_{\beta_{n,0}},$$

the oracle is represented as $f_{n,0} = \sum_{j=1}^{J} \beta_{n,0,j}\, \phi_j^*$, and it solves the score equation at $f_{n,0}$,

$$P_0 S_{f_{n,0}}(\phi_j^*) = 0 \;\; (\forall j).$$

Suppose $D_M^{(k)}(R_{n,0})$ has effective dimension $J_{n,0}$ with orthonormal basis $\phi^* \equiv \{\phi_j^*\}_{j=1}^{J_{n,0}}$, and we can project $f_n = \Pi(f_n \mid D_M^{(k)}(\mathcal{R}_{n,0}))$ from $D_M^{(k)}(R_{n,0})$ to $D_M^{(k)}(R_n)$, denoted by $\tilde{f}_n$. Since $\tilde{f}_n$ is an element on $D_M^{(k)}(\mathcal{R}_{n,0})$, we can also represent $\tilde{f}_n$ in terms of $\{\phi_j^*\}_{j=1}^{J_{n,0}}$, and we have $\tilde{f}_n = \sum_{j=1}^{J_{n,0}} \tilde{\beta}_n(j)\, \phi_j^*$.

S.4.1. *Outline.* We here give an outline of the proof first and then argue the rates of each terms along the process. Denote the score equation at f_n as $r_n(\cdot) = P_n S_{f_n}(\cdot)$, and denote the score equation at $\tilde{f}_n$ as $\tilde{r}_n(\cdot) = P_n S_{\tilde{f}_n}(\cdot)$. Define a linear operator T of an vector with J elements such that $T(b)(\cdot) = \sqrt{\frac{n}{J}} \sum_{j=1}^{J} b_j\, \phi_j^*(\cdot)$. Previously in Section 3.2 of the main paper, we define the normalized score vector at x by $\bar{\phi}_{n,x}(\cdot) \equiv J_{n,0}^{-1/2} \sum_{j \in \mathcal{R}_{n,0}} \phi_j^*(x)\phi_j^*(\cdot)$. Notice that $T(\phi^*(x))(\cdot) = \bar{\phi}_{n,x}$, and we shorthand it to be $\bar{\phi}$ in the following outline.

$$\begin{aligned}
&\sqrt{\frac{n}{J_{n,0}}}\big(\tilde{f}_n(x) - f_{n,0}(x)\big) \\
&\quad = T(\tilde{\beta}_n - \beta_{n,0})(x) \\
&\quad = T\Big(-\frac{\mathrm{d}}{\mathrm{d}\beta_{n,0}} P_0 S_{f_{n,0}}(\phi^*)\,(\tilde{\beta}_n - \beta_{n,0})\Big)(x) && \text{(Identity Outer Product)} \\
&\quad = T\big(-P_0\{S_{\tilde{f}_n} - S_{f_{n,0}}\}(\phi^*) + R_{1n}(\phi^*)\big)(x) && \text{(Taylor Expansion)} \\
&\quad = T\big(-P_0\{S_{\tilde{f}_n} - S_{f_n}\} - P_0\{S_{f_n} - S_{f_{n,0}}\} + R_{1n}\big)(\phi^*) && \text{(Add\&subtract } P_0 S_{f_n}) \\
&\quad = T\big(-\{P_{\tilde{f}_n} - P_{f_n}\} + (P_n - P_0)S_{f_n} - r_n + R_{1n}\big)(\phi^*) && \text{(Score expression)} \\
&\quad = -n^{1/2}\big((P_{\tilde{f}_n} - P_{f_n})(\bar{\phi})\big) + n^{1/2}(P_n - P_0)S_{f_{n,0}}(\bar{\phi}) + \\
&\qquad\quad (P_n - P_0)\big\{S_{f_n}(\bar{\phi}) - S_{f_{n,0}}(\bar{\phi})\big\} - n^{1/2} r_n(\bar{\phi}) + n^{1/2} R_{1n}(\bar{\phi}) && \text{(Linearity of } T) \\
&\quad = -o_P(1) + n^{1/2}(P_n - P_0)S_{f_{n,0}}(\bar{\phi}) + 0 \\
&\qquad\quad - o_P(1) + o_P(1) && \text{(Rate substitution)} \\
&\quad = n^{1/2}(P_n - P_0)S_{f_{n,0}}(\bar{\phi}) + o_P(1).
\end{aligned}$$

$\sqrt{J_{n,0}} S_{f_{n,0}}(\bar{\phi})$ serves as the influence curve of HAL-MLE, and it is a function of x since $\bar{\phi}$ is a function of x. For an application of the CLT we need that the variance under P_0 of $S_{\beta_{n,0}}(\bar{\phi})$ is $O(1)$. We have

$$S_{\beta_{n,0}}(\bar{\phi}) = \bar{\phi}(X) - P_{\beta_{n,0}}\bar{\phi}.$$

Therefore this variance can be bounded by $\|\bar{\phi}\|^2_{P_0}$. Recall the Positivity Condition for Theorem 4.3 of the main paper: the density p_f is bounded away from zero uniformly for all $f \in \mathcal{F}$. And the BSVN condition implies that the density p_f is uniformly bounded from above by some constant as well, and then this should generally be true for the oracle MLE. Then, it follows that

$$\|\bar{\phi}\|_{P_0} = O(\|\bar{\phi}\|_{P_{\beta_{n,0}}}) = O(J_{n,0}^{-1} \sum_{j \in \mathcal{R}_{n,0}} \{\phi_j^*(x)\}^2) = O(1).$$

Let $\sigma^2_{n,0} \equiv P_0\{S_{\beta_{n,0}}(\bar{\phi})\}^2$. We can then apply the empirical process theory here because $S_{f_{n,0}}(\bar{\phi})$ is independent from the sample data based on our construction of oracle MLE.

Then the rest of the proof of the pointwise asymptotic normality from the projection $\tilde{f}_n$ to the oracle $f_{n,0}$ depends on the rates of the terms $n^{1/2}\big(P_{\tilde{f}_n} - P_{f_n})(\bar{\phi})\big)$, $n^{1/2} r_n(\bar{\phi})$, and $n^{1/2} R_{1n}(\bar{\phi})$. By Appendix S.3, we have that $n^{1/2} r_n(\bar{\phi})$ is $o_P(1)$. For the rest of the terms, we will discuss them in the following subsections.

We want the property from HAL-MLE f_n to the truth f_0, thus we need to deal with the projection error from f_n to $\tilde{f}_n$ and the oracle approximation error from $f_{n,0}$ to f_0. And both of the errors could be bounded by the Uniform Approximation Assumption in Theorem 4.4 of the main paper for this working model with respect to $D_U^{(k)}([0,1])$. Since $f_n \in D_U^{(k)}([0,1])$, we certainly have that $\|f_n - \tilde{f}_n\|_\infty = O^+(1/J_{n,0}^{k+1})$. And $\|f_{n,0} - f_0\|_\infty = O^+(J_{n,0}^{-(k+1)})$. They become negligible when choosing $(n/J_{n,0})^{1/2} \asymp^+ J_{n,0}^{(k+1)}$. By adding these two terms, we achieve the pointwise asymptotic normality of HAL-MLE f_n to the truth f_0, as follows:

$$\sqrt{\frac{n}{J_{n,0}}}\big(f_n - f_{n,0}\big)(x) = -\,n^{1/2}(P_n - P_0) S_{f_{n,0}}(\bar{\phi}) + o_P(1) + \sqrt{\frac{n}{J_{n,0}}} O^+\big(J_{n,0}^{-(k+1)}\big),$$

and thus,

$$\begin{aligned}\sqrt{\frac{n}{J_{n,0}}}\big(f_n - f_0\big)(x) &= -\,n^{1/2}(P_n - P_0) S_{f_{n,0}}(\bar{\phi}) + o_P(1) + \sqrt{\frac{n}{J_{n,0}}} O^+\big(J_{n,0}^{-(k+1)}\big) \\ &= -\,n^{1/2}(P_n - P_0) S_{f_{n,0}}(\bar{\phi}) + o_P(1), \qquad \text{(S.4.1)}\end{aligned}$$

or equivalently, $\sigma_{n,0}^{-1} n^{1/2}(P_n - P_0) S_{\beta_{n,0}}(\bar{\phi}_{n,x})$ converges in distribution to $N(0,1)$, while $\sigma^2_{n,0} = O(1)$.

The Term of $R_{1n}(\bar{\phi})$. Here, we argue why $n^{1/2} R_{1n}(\bar{\phi})$ is $o_P(1)$. We get the $R_{1n}(\phi_j^*)$ term as a function of some basis function ϕ_j^* by the Taylor expansion of the score function $S_{f_n}(\phi_j^*)$ around $f_{n,0}$.

$$\begin{aligned}R_{1n}(\phi_j^*) &= P_0\{S_{\tilde{\beta}_n}(\phi_j^*) - S_{\beta_{n,0}}(\phi_j^*)\} - \frac{\mathrm{d}}{\mathrm{d}\beta_{n,0}} P_{\beta_{n,0}}(\phi_j^*)(\tilde{\beta}_n - \beta_{n,0}) \\ &= P_{\tilde{\beta}_n}(\phi_j^*) - P_{\beta_{n,0}}(\phi_j^*) - \frac{\mathrm{d}}{\mathrm{d}\beta_{n,0}} P_{\beta_{n,0}}(\phi_j^*)(\tilde{\beta}_n - \beta_{n,0}) \\ &= -\int \phi_j^*(p_{\tilde{\beta}_n} - p_{\beta_{n,0}}) - \frac{\mathrm{d}}{\mathrm{d}\beta_{n,0}} p_{\beta_{n,0}}(\tilde{\beta}_n - \beta_{n,0})\mathrm{d}x. \qquad \text{(S.4.2)}\end{aligned}$$

Denoting $R_{1n}(\tilde{f}_n, f_{n,0}) \equiv p_{\tilde{\beta}_n} - p_{\beta_{n,0}} - \frac{\mathrm{d}}{\mathrm{d}\beta_{n,0}} p_{\beta_{n,0}}(\tilde{\beta}_n - \beta_{n,0})$, and using the shorthanded notation $C(\tilde{f}_n) = C_n$ and $C(f_{n,0}) = C_{n,0}$ for the normalizing constant, then we have the following lemma.

LEMMA S.4.1.

$$\begin{aligned}R_{1n}(\phi_j) &= -\int \phi_j R_{1n}(\tilde{f}_n, f_{n,0})\mathrm{d}x \\ &= -C_{n,0}^{-2}\int \phi_j \exp(f_{n,0})\mathrm{d}x \cdot \tfrac{1}{2}\int \exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2\mathrm{d}x \\ &\quad + (C_n - C_{n,0})^2/(C_n C_{n,0}^2)\int \phi_j \exp(f_{n,0})\mathrm{d}x \\ &\quad + C_{n,0}^{-1}\tfrac{1}{2}\int \phi_j \exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2\mathrm{d}x \\ &\quad - (C_n - C_{n,0})/(C_n C_{n,0})\int \phi_j(\exp(\tilde{f}_n) - \exp(f_{n,0}))\mathrm{d}x.\end{aligned}$$

Proof: Let $C_n = C_n$ and $C_{n,0} = C_{n,0}$. We have

$$\begin{aligned}p_{\tilde{f}_n} - p_{f_{n,0}} &= C_n^{-1}\exp(\tilde{f}_n) - C_{n,0}^{-1}\exp(f_{n,0}) \\ &= \{C_n^{-1} - C_{n,0}^{-1}\}\exp(f_{n,0}) + C_n^{-1}(\exp(\tilde{f}_n) - \exp(f_{n,0})) \\ &= (C_{n,0} - C_n)/(C_n C_{n,0})\exp(f_{n,0}) + C_n^{-1}(\exp(\tilde{f}_n) - \exp(f_{n,0})) \\ &= -(C_n - C_{n,0})/C_{n,0}^2\exp(f_{n,0}) + (C_n - C_{n,0})^2/(C_n C_{n,0}^2)\exp(f_{n,0}) \\ &\quad + C_{n,0}^{-1}(\exp(\tilde{f}_n) - \exp(f_{n,0})) - (C_n - C_{n,0})/(C_n C_{n,0})(\exp(\tilde{f}_n) - \exp(f_{n,0})).\end{aligned}$$

By exact Taylor expansion of $\exp(x) = \exp(x_0) + \exp(x_0)(x - x_0) + \frac{1}{2}\exp(\xi(x, x_0))(x - x_0)^2$ for a $\xi(x, x_0)$ in between x and x_0, we have

$$\exp(\tilde{f}_n) - \exp(f_{n,0}) = \exp(f_{n,0})(\tilde{f}_n - f_{n,0}) + \tfrac{1}{2}\exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2.$$

We have

$$\begin{aligned}C_n - C_{n,0} &= \int(\exp(\tilde{f}_n) - \exp(f_{n,0}))\mathrm{d}x \\ &= \tfrac{1}{2}\int \exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2\mathrm{d}x + \int \exp(f_{n,0})(\tilde{f}_n - f_{n,0})\mathrm{d}x. \qquad \text{(S.4.3)}\end{aligned}$$

Thus,

$$\begin{aligned}-C_{n,0}^{-2}\exp(f_{n,0})(C_n - C_{n,0}) &= -C_{n,0}^{-2}\exp(f_{n,0})\int \exp(f_{n,0})(\tilde{f}_n - f_{n,0})\mathrm{d}x \\ &\quad - C_{n,0}^{-2}\exp(f_{n,0})\tfrac{1}{2}\int \exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2\mathrm{d}x.\end{aligned}$$

So

$$\begin{aligned}p_{\tilde{f}_n} - p_{f_{n,0}} &= -C_{n,0}^{-2}\exp(f_{n,0})\int \exp(f_{n,0})(\tilde{f}_n - f_{n,0})\mathrm{d}x \\ &\quad - C_{n,0}^{-2}\exp(f_{n,0})\tfrac{1}{2}\int \exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2\mathrm{d}x \\ &\quad + (C_n - C_{n,0})^2/(C_n C_{n,0}^2)\exp(f_{n,0}) + C_{n,0}^{-1}\exp(f_{n,0})(\tilde{f}_n - f_{n,0}) \\ &\quad + C_{n,0}^{-1}\tfrac{1}{2}\exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2 \\ &\quad - (C_n - C_{n,0})/(C_n C_{n,0})(\exp(\tilde{f}_n) - \exp(f_{n,0}))\end{aligned}$$

$$\equiv \frac{\mathrm{d}}{\mathrm{d}f_{n,0}} p_{f_{n,0}}(\tilde{f}_n - f_{n,0}) + R_{1n}(\tilde{f}_n, f_{n,0}),$$

where

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}f_{n,0}} p_{f_{n,0}}(\tilde{f}_n - f_{n,0}) &= -C_{n,0}^{-2}\exp(f_{n,0})\int \exp(f_{n,0})(\tilde{f}_n - f_{n,0})\mathrm{d}x \\
&\quad + C_{n,0}^{-1}\exp(f_{n,0})(\tilde{f}_n - f_{n,0}). \\
R_{1n}(\tilde{f}_n, f_{n,0}) &= -C_{n,0}^{-2}\exp(f_{n,0})\tfrac{1}{2}\int \exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2\mathrm{d}x \\
&\quad + (C_n - C_{n,0})^2/(C_n C_{n,0}^2)\exp(f_{n,0}) \\
&\quad + C_{n,0}^{-1}\tfrac{1}{2}\exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})^2 \\
&\quad - (C_n - C_{n,0})/(C_n C_{n,0})(\exp(\tilde{f}_n) - \exp(f_{n,0})).
\end{aligned}$$

This completes the proof of Lemma S.4.1.

We apply Lemma S.4.1 to the term $R_{1n}(\bar{\phi})$ and bound each term separately.

$$\begin{aligned}
R_{1n}(\bar{\phi}) &= -\int \bar{\phi}\left\{p_{f_n} - p_{f_{n,0}} - \frac{\mathrm{d}}{\mathrm{d}f_{n,0}} p_{f_{n,0}}(f_n - f_{n,0})\right\}\mathrm{d}x \\
&\equiv -\tfrac{1}{2}C_{n,0}^{-2}\int \bar{\phi}\exp(f_{n,0})\,\mathrm{d}x\int \exp\big(\xi(\tilde{f}_n, f_{n,0})\big)(\tilde{f}_n - f_{n,0})^2\,\mathrm{d}x \\
&\quad + \frac{(C_n - C_{n,0})^2}{C_n C_{n,0}^2}\int \bar{\phi}\exp(f_{n,0})\,\mathrm{d}x \\
&\quad + \tfrac{1}{2}C_{n,0}^{-1}\int \bar{\phi}\exp\big(\xi(\tilde{f}_n, f_{n,0})\big)(\tilde{f}_n - f_{n,0})^2\,\mathrm{d}x \\
&\quad - \frac{C_n - C_{n,0}}{C_n C_{n,0}}\int \bar{\phi}\big(\exp(\tilde{f}_n) - \exp(f_{n,0})\big)\,\mathrm{d}x.
\end{aligned}$$

For the first term, we know that $\exp\big(\xi(\tilde{f}_n, f_{n,0})\big)$ is bounded above and below, so we can treat it as a constant, and then we have

$$\begin{aligned}
C_{n,0}^{-2}\int \bar{\phi}\exp(f_{n,0})\,\mathrm{d}x\int \exp\big(\xi(\tilde{f}_n, f_{n,0})\big)(\tilde{f}_n - f_{n,0})^2\,\mathrm{d}x &\lesssim (P_{f_{n,0}}(\bar{\phi}))(\|\tilde{f}_n - f_{n,0}\|_\mu^2) \\
&= O(1)O_p(n^{-\frac{2k+2}{2k+3}}) = o_p(n^{-1/2})
\end{aligned}$$

We use the shorthand notation for $\|\cdot\|_{L^2(P_{f_{n,0}})}$ as $\|\cdot\|_{f_{n,0}}$. By Taylor expanding of $C(\tilde{f}_n)$ at $C_{n,0}$ as in Eq S.4.3, we have $C(\tilde{f}_n) - C_{n,0} = O_p(\|\tilde{f}_n - f_{n,0}\|_\mu)$. Then the second term can be bounded by $O(\|\phi\|_{f_{n,0}})$ times $\|\tilde{f}_n - f_{n,0}\|_\mu^2$.

$$\begin{aligned}
\frac{(C_n - C_{n,0})^2}{C_n C_{n,0}^2}\int \bar{\phi}\exp(f_{n,0})\,\mathrm{d}x &= \frac{(C_n - C_{n,0})^2}{C_n C_{n,0}}\int \frac{\bar{\phi}\exp(f_{n,0})}{C_{n,0}}\mathrm{d}x \\
&\lesssim (\|\tilde{f}_n - f_{n,0}\|_\mu^2)(\|\phi\|_{f_{n,0}}) \\
&= O_p(n^{-\frac{2k+2}{2k+3}})O(1) = o_p(n^{-1/2})
\end{aligned}$$

For the third term, again treating $\exp\big(\xi(\tilde{f}_n, f_{n,0})\big)$ as a constant, we need the Bounded Basis Assumption and Holder's inequality, and then we have,

$$\begin{aligned}\tfrac{1}{2}C_{n,0}^{-1}\int \bar{\phi}\exp\big(\xi(\tilde{f}_n, f_{n,0})\big)(\tilde{f}_n - f_{n,0})^2\,\mathrm{d}x &\lesssim \|\bar{\phi}\|_\infty \int (\tilde{f}_n - f_{n,0})^2\mathrm{d}x \\ &= O(J_{n,0}^{1/2})\,\|\tilde{f}_n - f_{n,0}\|_\mu^2 \\ &= O_p(n^{1/4k+6})O_p(n^{-\frac{2k+2}{2k+3}}) = o_p(n^{-1/2}),\end{aligned}$$

For the last term, we can similarly Taylor expand $\exp(\tilde{f}_n)$ on $\exp(f_{n,0})$, we have $\exp(\tilde{f}_n) - \exp(f_{n,0}) = \exp(\xi(\tilde{f}_n, f_{n,0}))(\tilde{f}_n - f_{n,0})$ for some ξ. Again using the Taylor Expansion of Eq S.4.3, and Cauchy-Schwartz inequality,

$$\begin{aligned}\frac{C_n - C_{n,0}}{C_nC_{n,0}}\int \bar{\phi}\big(\exp(\tilde{f}_n) - \exp(f_{n,0})\big)\,\mathrm{d}x &\lesssim \|\tilde{f}_n - f_{n,0}\|_\mu\|\phi\|_\mu\|\tilde{f}_n - f_{n,0}\|_\mu \\ &= O_p(n^{-\frac{k+1}{2k+3}})\,O(1)\,O_p(n^{-\frac{k+1}{2k+3}}) = o_p(n^{-1/2}).\end{aligned}$$

Thus, combining all four terms, we showed $n^{1/2}R_{1n}(\bar{\phi})$ is $o_P(1)$.

The Term $(P_{\tilde{f}_n} - P_{f_n})(\bar{\phi})$. Here, we argue why $n^{1/2}\big((P_{\tilde{f}_n} - P_{f_n})(\bar{\phi})\big)$ is $o_P(1)$. This term denotes the projection error of the score function $S_{f_n}(\bar{\phi})$ to the oracle score function $S_{f_{n,0}}(\bar{\phi})$. And we bound it in a slighly different way.

$$\begin{aligned}(P_{\tilde{f}_n} - P_{f_n})(\bar{\phi}) &= J_{n,0}^{-\frac{1}{2}}\sum_{j\in\mathcal{R}_{n,0}}(P_{\tilde{f}_n} - P_{f_n})(\phi_j^*)\phi_j^*(x) \\ &= J_{n,0}^{-\frac{1}{2}}\sum_{j\in\mathcal{R}_{n,0}}P_{f_{n,0}}(p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}})\phi_j^*\phi_j^*(x) \\ &= J_{n,0}^{-\frac{1}{2}}\sum_{j\in\mathcal{R}_{n,0}}P_{f_{n,0}}(p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}})\{\phi_j^* - P_{f_{n,0}}(\phi_j^*)\}\phi_j^*(x) \\ &= J_{n,0}^{-\frac{1}{2}}\sum_{j\in\mathcal{R}_{n,0}}\langle(p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}),(\phi_j^*)\rangle_{\beta_{n,0}}\phi_j^*(x) \\ &= J_{n,0}^{-\frac{1}{2}}\Pi(p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}\mid D^{(k)}(\mathcal{R}_{n,0}))(x)\end{aligned}$$

Notice that $p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}$ is a score in $L_0^2(P_{f_{n,0}})$, that is to say, the integration under measure $P_{f_{n,0}}$ is 0. And then use the inner product defined in Eq. (4.2) of the main paper, we can identify the term as a projection of a score in $L_0^2(P_{f_{n,0}})$ to the space $D^{(k)}(\mathcal{R}_{n,0})$.

$$\begin{aligned}\Pi(p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}\mid D^{(k)}(\mathcal{R}_{n,0}))(x) &\le \|\Pi(p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}\mid D^{(k)}(\mathcal{R}_{n,0}))\|_\infty \\ &\le \|p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}\|_\infty + \\ &\quad \|\Pi(p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}\mid D^{(k)}(\mathcal{R}_{n,0})) - (p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}})\|_\infty \\ &= O^+(1/J_{n,0}^{k+1}) + O^+(J_{n,0}^{-(k+1)}) \\ &= O^+(1/J_{n,0}^{k+1}).\end{aligned}$$

All the three densities $p_{\tilde{f}_n}$, p_{f_n}, and $p_{f_{n,0}}$ are bounded from above and below, thus $\|p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}\|_\infty = O(\|\tilde{f}_n - f_{n,0}\|_\infty) = O^+(J_{n,0}^{-(k+1)})$, by the Uniform Approximation

Assumption. Notice that $p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}}$ has bounded sectional variation as well, and thus it is in $D_U^{(k)}([0,1])$. Thus, we can apply the Uniform Approximation Assumption again to get $\|\Pi(p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}} \mid D^{(k)}(\mathcal{R}_{n,0})) - (p_{\tilde{f}_n} - p_{f_n}/p_{f_{n,0}})\|_\infty = O^+(J_{n,0}^{-(k+1)})$. We can do this because $D^{(k)}(\mathcal{R}_{n,0})$ is a close linear subspace equipped with the $L^2(P_{f_{n,0}})$ norm, and thus the projection is unique. So the projection could be difined as the limiting result of a universal steepest descent algorithm, proving that projection has distance of the same order.

Thus, $(P_{\tilde{f}_n} - P_{f_n})(\bar{\phi}) \asymp J_{n,0}^{(-1/2)} O^+(J_{n,0}^{-(k+1)}) = O_P^+(n^{-\frac{1}{2}}) = o_p(n^{-\frac{1}{2}})$, with some undersmoothening.

The Term $(P_n - P_0)\{S_{f_n}(\bar{\phi}) - S_{f_{n,0}}(\bar{\phi})\}$. Here, we argue why $(P_n - P_0)\{S_{f_n}(\bar{\phi}) - S_{f_{n,0}}(\bar{\phi})\} = 0$.

$$E_n(\bar{\phi}_{n,x}) \equiv (P_n - P_0)(S_{f_n}(\bar{\phi}_{n,x}) - S_{f_{n,0}}(\bar{\phi}_{n,x})).$$

Again, based on our score expression, we have that $S_{f_n}(\phi) - S_{f_{n,0}}(\phi) = (P_{f_{n,0}} - P_{f_n})\phi$ is a constant due to cancellation of ϕ. Therefore, the empirical process of a constant implies that $E_n(\bar{\phi}_{n,x}) = 0$.

□

S.4.2. *Proof of Corollary 4.7 in the main paper.*

PROOF. Let $g(f))(x) = \log p_f(x)$. Then, $g(f_0)(x) = \log p_0(x) = \log p_{f_0}(x)$. We wish to also analyze $g(f_n)(x) - g(f_0)(x)$ which implies the analysis for the density itself $p_{f_n}(x) - p_{f_0}(x)$ by a simple delta method argument. We note that

$$g(f_n)(x) - g(f_0)(x) = (f_n(x) - f_0(x)) - (\log C(f_n) - \log C(f_0)).$$

We have a linear approximation for $f_n(x) - f_0(x)$ given in Theorem 4.4 of the main paper, namely $J_{n,0}^{1/2}(P_n - P_0)S_{f_{n,0}}(\bar{\phi}_{n,x})$. We can Taylor expand

$$\log C(f_n) - \log C(f_0) = \frac{1}{C(f_0)}(C(f_n) - C(f_0)) - \frac{1}{C(\xi)^2}(C(f_n) - C(f_0))^2,$$

where ξ is between f_n and f_0. Knowing $C(\xi)$ to be bounded,

$$(C(f_n) - C(f_0))^2 = (\int_0^1 \exp(f_n - f_0)dx)^2 = (\int_0^1 \exp(\xi_2)(f_n - f_0)dx)^2 = O_p(\|f_n - f_0\|_\mu^2),$$

by another Taylor expansion with ξ_2 and $\exp(\xi_2)$ to be bounded. Thus, we have the second order remainder to be negligible by $(n/J_{n,0})^{1/2} O_p(\|f_n - f_0\|_\mu^2) = o_p(1)$.

Then, notice that $C(f_0)$ is a pathwise differentiable function of f_0, choosing a path $f_\epsilon^h = f + \epsilon h$ where $\mathbb{E}_{P_{f_n}}[h] = 0$,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\varepsilon} C(f_\varepsilon^h)\Big|_{\varepsilon=0} &= \int_0^1 e^{f_0(x)} h(x)\,\mathrm{d}x \\ &= \int_0^1 C(f)\, h\, \mathrm{d}P_f = 0 \end{aligned}$$

This implies $D_f^* = 0$ for $C(f)$, i.e., the normalizing constant only changes in second order as one changes P_f. It will not contribute to linearization and we can treat $g(f_n)(x) - g(f_0)(x)$ as $f_n(x) - f_0(x)$.

Therefore, we can conclude that

$$(n/J_{n,0})^{1/2}(g(f_n)(x)-g(f_0)(x))=(n/J_{n,0})^{1/2}(f_n(x)-f_0(x))+o_P(1).$$

Therefore, $\log p_{f_n}(x)-\log p_{f_0}(x)$ behaves as $(f_n-f_0)(x)$, which proves that

$$(n/J_{n,0})^{1/2}(p_{f_n}-p_{f_0})(x)=p_0(x)(n/J_{n,0})^{1/2}(f_n-f_0)(x)+o_P(1).$$

□

S.5. Proof of Uniform Convergence. This appendix includes the proof of Theorem 4.8 in the main paper.

PROOF. We extended the Basis Boundedness Assumption in Theorem 4.6 of the main paper such that $R_{1n}(\bar{\phi}_{n,x})$ and $\tilde{r}_n(\bar{\phi}_{n,x})$ hold uniformly in $x\in[0,1]$. The uniform convergence result is obtained by using empirical process bounds in terms of the entropy integral to the empirical process $(n^{1/2}(P_n-P_0)S : S\in\mathcal{S}_n)$ with $\mathcal{S}_n=\{S_{f_{n,0}}(\bar{\phi}_{n,x}) : x\in[0,1]\}$. Notice that, by orthonormality and boundedness,

$$\|J_{n,0}^{-\frac{1}{2}}\bar{\phi}_{n,x}\|^2_{L^2(P_{f_{n,0}})}=J_{n,0}^{-2}\sum_{j\in\mathcal{R}_{n,0}}\phi_j^*(x)^2\;\lesssim\;J_{n,0}^{-1},$$

hence we may take $\|J_{n,0}^{-\frac{1}{2}}S_{f_{n,0}}(\bar{\phi}_{n,x})\|_{L^2(P)}\lesssim J_{n,0}^{-1/2}$. That is to say, the L^2-norm of these functions shrinks to zero at the rate $J_{n,0}^{-1/2}$. Also, because $x\mapsto\bar{\phi}_{n,x}$ is a one–dimensional mapping of x, its covering numbers satisfy

$$\log N\big(\epsilon,\;\mathcal{S}_n,\;L^2(P_{f_{n,0}})\big)\;\lesssim\;\log(1/\epsilon),$$

uniformly in n. Consequently, the entropy integral obeys the VC-1 form

$$J(\delta,\mathcal{S}_n,L^2(P_{f_{n,0}}))\;\equiv\;\int_0^\delta\sqrt{\log(1/\epsilon)}\,\mathrm{d}\varepsilon\;\asymp\delta\sqrt{\log(1/\delta)}.$$

So, let function class $\mathcal{F}_{S_n}$ be the function class that shrinks every function $s\in\mathcal{S}_n$ by $J_{n,0}^{\frac{1}{2}}$. Then this function class $\mathcal{F}_{S_n}$ has a shrinkage L^2-norm and an entropy integral $J_2(\delta,\mathcal{S}_n)$

Recall that

$$\sqrt{\frac{n}{J_{n,0}}}\big(f_n-f_0\big)(x)=-\,n^{1/2}(P_n-P_0)S_{f_{n,0}}(\bar{\phi})+o_P(1).$$

Then, with a similar approach of bounding the empirical process in Appendix S.2,

$$\begin{aligned}\sup_{x\in[0,1]}|\,n^{1/2}(P_n-P_0)S_{f_{n,0}}(\bar{\phi})|&=J_{n,0}^{\frac{1}{2}}\sup_{x\in[0,1]}|n^{1/2}(P_n-P_0)\Big(\frac{S_{f_{n,0}}(\bar{\phi})}{J_{n,0}^{\frac{1}{2}}}\Big)|\\&=J_{n,0}^{\frac{1}{2}}\sup_{f\in\mathcal{F}_{S_n},\|f\|_2<\delta}|\mathbb{G}_n(f)|\\&\lesssim J_{n,0}^{\frac{1}{2}}J_2(\delta_n,\mathcal{F}_{S_n}),\text{with a }\delta_n\text{ that makes }J_2(\delta_n)\text{ dominate}\\&=J_{n,0}^{\frac{1}{2}}\delta_n\sqrt{\log(1/\delta_n)}.\end{aligned}$$

We know that $f\in\mathcal{F}_{S_n},\|f\|_2<J_{n,0}^{-1/2}$, then we can choose $\delta_n\asymp J_{n,0}^{-1/2}$. First, it is a valid choice for $J_2(\delta)$ to dominate, because $J_2(\delta_n)\asymp\delta_n\log(1/\delta_n)\lesssim\delta_n^2n^{1/2}$. Then, $J_{n,0}^{\frac{1}{2}}J_2(\delta_n,\mathcal{F}_{S_n})=\sqrt{\log(1/\delta_n)}\asymp\sqrt{\log n}$.

With $J_{n,0} \asymp^+ n^{1/(2k+3)}$ this reads

$$\sup_{x\in[0,1]} \left|\sqrt{n}(P_n - P_0)S_{f_{n,0}}(\bar{\phi}_{n,x})\right| \;=\; O_p\Big(\sqrt{\log n}\Big).$$

Then it follows:

$$(n/J_{n,0})^{1/2}\|f_n - f_0\|_\infty = O_P(\sqrt{\log n}),$$

so that $\|f_n - f_0\|_\infty = O_P^+(n^{-\frac{(k+1)}{(2k+3)}})$.

$\square$

S.6. Proof of Asymptotic Efficiency. This appendix includes the proof of Theorem 5.1 and Theorem 5.2 in the main paper. We provide a standard TMLE proof establishing asymptotic efficiency under the assumption that $D_M^{(k)}(\mathcal{R}_n)$ approximates $D_U^{(k)}([0,1])$.

S.6.1. *Proof of Theorem 5.1 in the main paper.*

PROOF. We have that $P_n S_{f_n}(\phi_j) = 0$ for the $J_n - 1$ scores that define the L_1 constrained score space in $D^{(k)}(\mathcal{R}_n)$. The tangent space generated by paths through f is given by closure of linear span $\{S_f(\phi) : \phi \in D_U^{(k)}([0,1])\}$. Thus, we can represent $D_f^* = S_f(\phi_f)$ for some $\phi_f \in D_U^{(k)}([0,1])$. Therefore, if the basis $\mathcal{R}_n$ is chosen so that $D^{(k)}(\mathcal{R}_n)$ yields a $O^+(1/J_n^{k+1})$ L^2-approximation of $D_U^{(k)}([0,1])$ and this L^2 approximation can be extended to the corresponding L_1 constrained score space, then we can find a projection $\tilde{\phi}_f$ of ϕ_f onto the L_1 constrained score space of $D^{(k)}(\mathcal{R}_n)$ that approximates ϕ_f in L^2-norm within distance $O^+(1/J_n^{k+1})$. Therefore, following the same argument as in Appendix S.3, we have

$$\begin{aligned}
P_n D_{f_n}^* = P_n S_{f_n}(\phi_{f_n}) &= P_n\{S_{f_n}(\phi_{f_n}) - S_{f_n}(\tilde{\phi}_{f_n})\} + P_n S_{f_n}(\tilde{\phi}_{f_n}) \\
&= (P_n - P_0)\{\phi_{f_n} - \tilde{\phi}_{f_n}\} + \\
&\quad\; (P_0 - P_{f_n})\{\phi_{f_n} - \tilde{\phi}_{f_n}\} + 0,
\end{aligned}$$

where we note that the expectation of scores at f_n under P_{f_n} equals zero, because $S_{f_n}(\tilde{\phi}_{f_n}) = \tilde{\phi}_{f_n} - P_{f_n}\tilde{\phi}_{f_n}$ by definition.

S.6.1.0.1. Bounding the First Empirical Process Term. We could directly use that the covering number of the basis class is the covering number of $D_U^{(k)}([0,1])$. And using the L^2 covering number from Appendix S.2, we can bound the first term can be bounded by $n^{-1/2}J_2(\delta_n, D_U^{(k)}([0,1]))$ with $\delta_n \asymp n^{-\frac{k+1}{2k+3}}$, where $J_2(\delta, D_U^{(k)}([0,1]) \asymp \delta^{(2k+1)/(2k+2)}$. Thus, we have

$$\begin{aligned}
\left|(P_n - P_0)\left\{\phi_{f_n} - \tilde{\phi}_{f_n}\right\}\right| &= O_P\left(n^{-1/2}J_2(\delta_n, D_U^{(k)}([0,1]))\right) \\
&= O_P\left(n^{-1/2}\delta_n^{\frac{2k+1}{2k+2}}\right) \\
&= O_P\left(n^{-\frac{2k+2}{2k+3}}\right) \\
&= o_P\left(n^{-1/2}\right).
\end{aligned}$$

S.6.1.0.2. Bounding the Second Term. The second term can be bounded with Cauchy-Schwarz inequality by $\|p_{f_n} - p_0\|_\mu \|\phi_{f_n} - \tilde{\phi}_{f_n}\|_\mu$, where we shorthanded $\|\cdot\|_{L^2(\mu)}$ by $\|\cdot\|_\mu$. By the L^2 convergence result in Theorem 4.3 of the main paper, we have $\|p_{f_n} - p_0\|_\mu = O_p(n^{-(k+1)/(2k+3)})$. By the L^2 Approximation Assumption, $\|\phi_{f_n} - \tilde{\phi}_{f_n}\|_\mu = O^+(J_n^{-(k+1)})$. So that we obtain the bound $O_p^+(n^{-(k+1)/(2k+3)} J_n^{-(k+1)})$. Choose $J_n \asymp^+ n^{1/(2k+3)}$, which could be achieved by cross validation or slight undersmoothening. Again, it follows that for all k, the second term is $o_P(n^{-1/2})$.

$$\begin{aligned}
\left|(P_0 - P_{f_n})\left\{\phi_{f_n} - \tilde{\phi}_{f_n}\right\}\right| &\le \|p_{f_n} - p_0\|_\mu \cdot \left\|\phi_{f_n} - \tilde{\phi}_{f_n}\right\|_\mu \\
&\le \|p_{f_n} - p_0\|_\mu \cdot \left\|\phi_{f_n} - \tilde{\phi}_{f_n}\right\|_\mu \\
&= O_P\left(n^{-\frac{k+1}{2k+3}}\right) \cdot O^+\left(J_n^{-(k+1)}\right) \\
&= O_P^+\left(n^{-\frac{k+1}{2k+3}} n^{-\frac{k+1}{2k+3}}\right) \\
&= o_P\left(n^{-1/2}\right),
\end{aligned}$$

Thus, we here finished proving Theorem 5.1 in the main paper. □

S.6.2. *Proof of Theorem 5.2 in the main paper.*

PROOF. Given we have established $P_n D^*_{f_n} = o_P(n^{-1/2})$, we have

$$\Psi^F(f_n) - \Psi^F(f_0) = (P_n - P_0) D^*_{f_n} + R(P_{f_n}, P_{f_0}) + o_p(n^{-1/2}).$$

The Positivity Condition implies that the rate of $d_0(f_n, f_0)$ has the same rate of convergence in $L^2(\mu)$-norm. Then, by the Negligible Remainder Assumption in Theorem 5.2 of the main paper, we can generally bound $R(P_{f_n}, P_{f_0})$ by $\|p_{f_n} - p_{f_0}\|^2_\mu$. So then $R(P_{f_n}, P_{f_0}) = O_P(n^{-\frac{2k+2}{2k+3}})$. We also have that $\{D_f : f \in D_U^{(k)}([0,1])\}$ is a P_0-Donsker class and we assume $P_0\{D^*_{f_n} - D^*_{f_0}\}^2 \to_p 0$.

Therefore,

$$(P_n - P_0)\{D^*_{f_0} - D^*_{f_n}\} = n^{-1/2}\, o_p(1) = o_p(n^{-1/2}).$$

This then proves

$$\Psi^F(f_n) - \Psi^F(f_0) = (P_n - P_0) D^*_{f_0} + o_P(n^{-1/2}).$$

This proves asymptotic efficiency of $\Psi(f_n)$ as an estimator of $\Psi(f_0)$. □

S.6.3. *Verification of the Assumptions for Median.*

PROOF. Recall that for the median functional $\psi(f) = F^{-1}(0.5)$, the efficient influence function is given by

$$D_f(X) = \frac{1}{f(\psi)}\left(\tfrac{1}{2} - I(X \le \psi)\right),$$

where f is the corresponding density and $\psi = \psi(f)$ is the median. Let $\psi_n = \psi(f_n)$ and $\psi_0 = \psi(f_0)$.

S.6.3.0.1. Verification of the Second Assumption. For the second assumption, we aim to show that

$$P_0\{D_{f_n} - D_{f_0}\}^2 = o_p(1).$$

We have $\|f_n - f_0\|_\infty = O_P^+(n^{-\frac{k+1}{2k+3}})$. Then the CDFs F_n and F_0 are close to each other in the sense that $\|F_n - F_0\|_\infty \le \|f_n - f_0\|_1 \le \|f_n - f_0\|_\infty = O_P^+(n^{-\frac{k+1}{2k+3}})$.

Since $F_n(\psi_n) = F_0(\psi_0) = \frac{1}{2}$, we have

$$0 = F_n(\psi_n) - F_0(\psi_0) = \{F_0(\psi_n) - F_0(\psi_0)\} + \{F_n(\psi_n) - F_0(\psi_n)\}.$$

By the mean value theorem, there exists some ξ_n between ψ_n and ψ_0 such that

$$F_0(\psi_n) - F_0(\psi_0) = f_0(\xi_n)\,(\psi_n - \psi_0).$$

Taking absolute values yields

$$|f_0(\xi_n)|\,|\psi_n - \psi_0| = |F_n(\psi_n) - F_0(\psi_n)|.$$

Since $|F_n(\psi_n) - F_0(\psi_n)| \le \|F_n - F_0\|_\infty = O_P^+\Big(n^{-\frac{k+1}{2k+3}}\Big)$ and $\|f_0\|_\infty = O(1)$, we obtain

$$|\psi_n - \psi_0| = O_P^+\Big(n^{-\frac{k+1}{2k+3}}\Big).$$

We start by expanding

$$\begin{aligned} D_{f_n} - D_{f_0} &= \frac{1}{f_n(\psi_n)}\big(\tfrac{1}{2} - I(X \le \psi_n)\big) - \frac{1}{f_0(\psi_0)}\big(\tfrac{1}{2} - I(X \le \psi_0)\big) \\ &= \frac{f_0(\psi_0) - f_n(\psi_n)}{f_n(\psi_n) f_0(\psi_0)}\big(\tfrac{1}{2} - I(X \le \psi_0)\big) + \frac{1}{f_n(\psi_n)}\big(I(X \le \psi_0) - I(X \le \psi_n)\big). \end{aligned}$$

If we bound the $L^2(P_0)$-norm for the two terms on the right hand side, then we bound the left hand side.

For the first term, write

$$f_0(\psi_0) - f_n(\psi_n) = \big(f_0(\psi_0) - f_n(\psi_0)\big) + \big(f_n(\psi_0) - f_n(\psi_n)\big).$$

Assuming f is differentiable with bounded derivative in a neighborhood of ψ_0, the mean-value theorem gives

$$f_n(\psi_0) - f_n(\psi_n) = -f_n^{(1)}(\xi_n)(\psi_n - \psi_0)$$

for some ξ_n between ψ_n and ψ_0. The $L^2(P_0)$-norm of the first term is bounded by $\|f_n - f_0\|_{P_0} + |\psi_n - \psi_0| = O_P^+(n^{-\frac{k+1}{2k+3}}) = o_P(1)$.

For the second term, note that WLOG, we have $\psi_0 < \psi_n$. Since $\psi_n - \psi_0$ converges to zero we have that the $L^2(P_0)$-norm of the latter indicator converges to zero.

S.6.3.0.2. Verification of the Third Assumption. We aim to show that

$$R(P_{f_n}, P_0) = o_p(n^{-1/2}).$$

$$\begin{aligned} R(P_{f_n}, P_0) &= \psi_n - \psi_0 + P_0 D_{f_n} \\ &= \psi_n - \psi_0 + \frac{F_0(\psi_0) - F_0(\psi_n)}{f_n(\psi_n)} \\ &= \psi_n - \psi_0 + \frac{f_0(\tilde{\xi}_n)}{f_n(\psi_n)}(\psi_0 - \psi_n) \\ &= \frac{f_n(\psi_n) - f_0(\tilde{\xi}_n)}{f_n(\psi_n)}(\psi_n - \psi_0) \\ &\le \|f_n - f_0\|_\infty \cdot |\psi_n - \psi_0| = O_P^+(n^{-\frac{k+1}{2k+3}}) \cdot O_P^+(n^{-\frac{k+1}{2k+3}}) = o_P(n^{-1/2}). \end{aligned}$$

□

S.6.4. *Proof of Theorem 5.3 in the main paper.* For a single-step HAL-TMLE, knowing that f_n^1 converges to f_0 at a faster rate than $n^{-1/4}$, we have $P_n D^*_{f_n^1} = o_p(n^{-1/2})$, according to van der Laan (2017). And thus the same proof of asymptotic efficiency holds for HAL-TMLE as well by extending all the f_n to f_n^1. This is why we claim the proof above to be a standard TMLE proof.

S.7. Simulation Data Generating Process Setup.

S.7.1. *Data Generating Process for Section 6 of the main paper.* We here provide the detailed data generating process (DGP) for the simulation study in Section 6 of the main paper. All distributions are defined on the support $[0,1]$ and we implement six different DGPs to evaluate the performance of our HAL-MLE methods across various distributional shapes and complexities.

S.7.1.1. *DGP Definitions and Parameters.*

S.7.1.1.1. 1. Truncated Normal Distribution. The truncated normal distribution is defined as:

$$f(x) = \frac{\phi\left(\frac{x-\mu}{\sigma}\right)}{\sigma\left[\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)\right]}, \quad x \in [a,b] \tag{S.7.1}$$

where ϕ and Φ are the standard normal PDF and CDF respectively.

Parameters: $\mu = 0.5$, $\sigma = 0.1$, $a = 0$, $b = 1$

S.7.1.1.2. 2. Sinusoidal Distribution. The sinusoidal-based density is defined as:

$$f(x) = \frac{\sin(\pi x) + 1.1}{C}, \quad x \in [0,1] \tag{S.7.2}$$

where $C = \int_0^1 (\sin(\pi t) + 1.1)dt = \frac{2}{\pi} + 1.1 \approx 1.7366$ is the normalization constant.

Parameters: No configurable parameters (fixed functional form)

S.7.1.1.3. 3. Truncated GMM Symmetric Three Components. A mixture of three truncated normal distributions:

$$f(x) = \sum_{i=1}^{3} w_i \cdot f_i(x), \quad x \in [0,1] \tag{S.7.3}$$

where each $f_i(x)$ is a truncated normal component.

Parameters:

- Component 1: $\mu_1 = 0.2$, $\sigma_1 = 0.05$, $w_1 = 0.33$
- Component 2: $\mu_2 = 0.5$, $\sigma_2 = 0.05$, $w_2 = 0.34$
- Component 3: $\mu_3 = 0.8$, $\sigma_3 = 0.05$, $w_3 = 0.33$

S.7.1.1.4. 4. Truncated GMM Five Spikes. A mixture of six truncated normal components with five narrow spikes and one broad component:

$$f(x) = \sum_{i=1}^{6} w_i \cdot f_i(x), \quad x \in [0,1] \tag{S.7.4}$$

Parameters:

- Spike components ($i = 1, \ldots, 5$): $\mu_i \in \{0.45, 0.475, 0.5, 0.525, 0.55\}$, $\sigma_i = 0.005$, $w_i = 1/15$
- Broad component: $\mu_6 = 0.5$, $\sigma_6 = 0.05$, $w_6 = 2/3$

S.7.1.1.5. 5. Truncated GMM Asymmetric Three Components. An asymmetric mixture of three truncated normal distributions:

$$f(x) = \sum_{i=1}^{3} w_i \cdot f_i(x), \quad x \in [0,1] \tag{S.7.5}$$

Parameters:

- Component 1: $\mu_1 = 0.35$, $\sigma_1 = 0.1$, $w_1 = 0.4$
- Component 2: $\mu_2 = 0.65$, $\sigma_2 = 0.05$, $w_2 = 0.4$
- Component 3: $\mu_3 = 0.9$, $\sigma_3 = 0.2$, $w_3 = 0.2$

S.7.1.1.6. 6. Step Function Distribution. A piecewise constant distribution:

$$f(x) = \frac{1}{C} \begin{cases} \ell_1, & x \in [0,b) \\ \ell_2, & x \in [b,1] \end{cases} \tag{S.7.6}$$

where $C = \ell_1 \cdot b + \ell_2 \cdot (1-b)$ is the normalization constant.

Parameters: $\ell_1 = 1.0$, $\ell_2 = 0.5$, $b = 0.7$

S.7.1.2. *Population Parameters.* Table 1 summarizes the true population parameters for each DGP, including the mean, median, variance, second moment, and survival probability at $x = 0.5$.

TABLE 1
Population Parameters for Data Generating Processes

DGP	Mean	Median	Variance	Second Moment	$S(0.5)$
Truncated Normal	0.5000*	0.5000*	0.010000	0.260000	0.5000*
Sinusoidal	0.5000*	0.5000*	0.070145	0.320145	0.5000*
GMM Symmetric	0.5000*	0.5000*	0.061896	0.311896	0.5000*
GMM Five Spikes	0.5000*	0.5000*	0.002092	0.252092	0.5000*
GMM Asymmetric	0.559669	0.608370	0.041087	0.354317	0.619610
Step Function	0.438235	0.425000	0.071283	0.263333	0.411760

*Exact values due to symmetry around $x = 0.5$
$S(0.5) = P(X > 0.5)$ is the survival probability at $x = 0.5$

The population parameters in Table 1 serve as ground truth for evaluating the accuracy of our HAL-MLE estimators across different distributional characteristics.

S.7.2. *Random Seed Generation.* In this subsection, we provide the random seeds used for generating the simulation data. A master seed ($m = 42$) generates a deterministic sequence of replicate seeds $\{s_r\}_{r=1}^{R}$ with $R = 1000$ per (DGP, n). In replicate r the data sampler is seeded by s_r; cross-validation fold assignments use the same s_r (ensuring identical partitions across estimators).

S.7.3. *Configuration of Cross-Validation with Optuna.* We use a 5-fold cross-validation to cross validate over the L_1 norm constraint and the basis order of the HAL-MLE. Instead of cross validating over each grid of the hyperparameter pairs, we used Optuna (Akiba et al., 2019) to sample the hyperparameters with 50 steps. Throughout the cross-validation steps, we acknowledge the failure of the solvers occasionally and we have to switch to other solvers in `CVXPY`. In general, `MOSEK` is the fastest solver, and we use it as the default solver. `SCS` is relatively slow but more robust. The search space is L_1 norm constraint in $[10^{-3}, 10^6]$ (log scale) and basis order in $\{0, 1, 2\}$. The trial budget is 50 evaluations per estimator/DGP/sample size. As claimed in (Akiba et al., 2019), the performance of Optuna is very close to the performance of the grid search with a massive reduction in computational cost.

S.8. Trend Filtering with a Specialized ADMM Algorithm for Density Estimation. In this appendix, we describe how we extend the Trend Filtering (TF) method to our density estimation framework. We start from the equivalency between zero-order falling factorial basis and zero-order truncated power basis. This allows us to reparameterize our zero-order HAL-MLE to TF with zero-order divided differences. We start with the likelihood function p_{fn} and construct the likelihood using linear combinations of basis functions through the link function. By reparameterizing our $\phi\beta$ as θ, we face the challenge of the normalizing constant $\int_0^1 \exp(\phi\beta)dx$, which we approximate by breaking the integral into a sum over grid bins, yielding the normalizing constant $\sum_j \exp(\theta_j)\Delta x_j$.

S.8.1. *Problem Formulation.*

S.8.1.1. *Uniform Grid Construction.* Our current implementation in `methods/non_HAL_method/TF_CVXPY` uses a *uniform* grid on $[0, 1]$ (rather than a data-adaptive grid). This avoids numerical pathologies caused by extremely small gaps in data-adaptive grids when n is large, and matches the grid choice used elsewhere in our experiments.

1. Choose the number of bins $J = n + 1$ (so the parameter dimension is comparable to the legacy data-adaptive choice).
2. Define uniform knots: $x_j = j/J$ for $j = 0, 1, \dots, J$.
3. Create bins: $[x_0, x_1), [x_1, x_2), \dots, [x_{J-1}, x_J]$ with widths $\Delta x_j = x_{j+1} - x_j = 1/J$.

Given observations $x_1, \dots, x_n$, let c_j be the histogram count in bin j:

$$c_j \;=\; \#\{i : \, x_i \in [x_j, x_{j+1})\}, \qquad j = 0, 1, \dots, J-1,$$

so that $\sum_{j=0}^{J-1} c_j = n$.

S.8.1.2. *Likelihood Construction and Reparameterization.* We start with the likelihood function p_{fn} and construct the likelihood using linear combinations of basis functions through the link function. The key insight is that the TF basis functions are equivalent to truncated power basis functions, allowing us to reparameterize our original $\phi\beta$ parameters as θ.

We model the log-density as piecewise constant over the grid:

$$\log p_\theta(x) = \theta_j, \quad x \in [x_j, x_{j+1}), \quad j = 0, 1, \dots, J-1.$$

As in the implementation, we fix an identifiability constraint $\theta_0 = 0$ and optimize over the full vector $\theta = (\theta_0, \dots, \theta_{J-1}) \in \mathbb{R}^J$ subject to this constraint.

The normalizing constant is approximated by a Riemann sum on the grid:

$$Z(\theta) = \sum_{j=0}^{J-1} \exp(\theta_j)\,\Delta x_j.$$

The corresponding density function is:

$$p_\theta(x) = \frac{\exp(\theta_j)}{Z(\theta)}, \quad x \in [x_j, x_{j+1}).$$

Using the binned likelihood (i.e., grouping observations by their bin counts), the negative log-likelihood becomes:

$$f(\theta) = -\sum_{j=0}^{J-1} c_j \theta_j + n \log Z(\theta).$$

S.8.1.3. *Total Variation Penalty.* The total variation representation of TF is the ℓ_1 norm of $\phi^{-1}\theta$. Since TF does not penalize the parametric part, this representation simplifies to the ℓ_1 norm of the zero-order divided difference of θ. We can further extend this to k-th order divided differences.

The k-th order difference matrix $D^{(k)}$ is constructed recursively by Eq(3) of (Ramdas and Tibshirani, 2016).

The complete optimization problem is:

$$\boxed{\min_{\theta \in \mathbb{R}^J} f(\theta) \quad \text{s.t.} \quad \theta_0 = 0,\ \|D^{(k+1)}\theta\|_1 \le C} \tag{S.8.1}$$

This constrained form is equivalent (under standard constraint qualification conditions) to a penalized form with a Lagrange multiplier $\lambda \ge 0$:

$$\min_{\theta:\, \theta_0 = 0} f(\theta) + \lambda \|D^{(k+1)}\theta\|_1.$$

S.8.2. *ADMM Algorithm.* Following the approach in the trend filtering literature, we adapt the specialized ADMM algorithm for trend filtering to our density estimation case. The algorithm still consists three steps.

S.8.2.1. *Augmented Lagrangian.* We introduce auxiliary variables $\alpha = D^{(k)}\theta$. And we use 0 as a placeholder for θ_0 and such that $\theta = (0, \tilde{\theta})$ is the full parameter vector. The normalization is achieved by $Z(\theta)$. The augmented Lagrangian is:

$$\begin{aligned}\mathcal{L}_\rho(\theta, \alpha, u) = f(\theta) &+ \lambda \|D^{(1)}\alpha\|_1 \\ &+ \frac{\rho}{2}\|\alpha - D^{(k)}\theta + u\|_2^2 - \frac{\lambda}{2}\|u\|_2^2\end{aligned}$$

where u is the scaled dual variable and in practice we choose $\rho = \lambda$ in practice Ramdas and Tibshirani (2016).

S.8.2.2. *ADMM Iterations.* The ADMM algorithm alternates between three updates:

S.8.2.2.1. θ-step (Non-smooth Optimization):.

$$\tilde{\theta}^{t+1} = \arg\min_{\tilde{\theta}} f(\tilde{\theta}) + \frac{\rho}{2}\|\alpha^t - D^{(k)}(0, \tilde{\theta}^t) + u^t\|_2^2$$

The first step no longer has a closed-form solution like in the regression case Ramdas and Tibshirani (2016) due to the likelihood. So we have to solve it using the L-BFGS algorithm.

S.8.2.2.2. α-step (Fused Lasso):.

$$\alpha^{t+1} = \arg\min_{\alpha} \frac{1}{2}\|\alpha - v^t\|_2^2 + \frac{\lambda}{\rho}\|D^{(1)}\alpha\|_1$$

where $v^t = D^{(k)}\theta^{t+1} - u^t$.

This is a standard fused lasso problem that can be solved using the solution path algorithm (Tibshirani, 2011).

S.8.2.2.3. Dual Update:.

$$u^{t+1} = u^t + \alpha^{t+1} - D^{(k)}(0, \tilde{\theta}^{t+1})$$

S.8.2.3. *Convergence Criteria.* We monitor primal and dual residuals:

$$r^t = \alpha^t - D^{(k)}\theta^t \quad \text{(primal residual)}$$

$$s^t = \rho D^{(k)T}(\alpha^t - \alpha^{t-1}) \quad \text{(dual residual)}$$

The algorithm terminates when:

$$\|r^t\|_2 \leq \varepsilon_{\text{pri}} \quad \text{and} \quad \|s^t\|_2 \leq \varepsilon_{\text{dual}}$$

with tolerances $\varepsilon_{\text{pri}} = 10^{-4}\sqrt{m}$ and $\varepsilon_{\text{dual}} = 10^{-4}\sqrt{n}$, where m is the dimension of α.

S.8.3. *TFPP.* We include the variant of TF with penalty on the parametric part as TFPP, by replacing the divided difference matrix $D^{(k)}$ with the variant H in Eq(10) and Eq(11) of Wang, Smola and Tibshirani (2014). And so others can be adapted accordingly. Notice that Eq(10) and Eq(11) of Wang, Smola and Tibshirani (2014) are for the data-adaptive knots, and we use the uniform knots here simply by replacing the diagonal matrix $\Delta^{(k)}$ with an scaled identity matrix, such that the ADMM algorithm can be adapted accordingly.

One problem of adapting this construction to data-adaptive knots is that the entries of the inverse of matrix H can be extremely large when we sample from the distribution defined on $[0, 1]$. The data points can get extremely close to each other which a gap of 10^{-16} or smaller. This can cause numerical instability when we iterate the $D^{(k)}$, and the optimization can diverge. We can moderate this problem by computing the inverse of matrix H using Wang, Smola and Tibshirani (2014, Algorithm 2). This algorithm is more stable because the matrix H, with entries of 10^{16} or larger, can be calculated implicitly. However, this algorithm suffers from the little gap between the data points as well. The imbalance weighting of the constraints can cause the optimization to diverge. A similar ADMM algorithm for TFPP density estimation with data-adaptive knots is still under development and will be addressed in the future.

Another problem is that this data-adaptive knots approach is only defined in the univariate case, since the order statistic used for constructing $\Delta^{(k)}$ is not defined in the multivariate case. The paper Sadhanala et al. (2024), as a multivariate extension of TF, only considers the exponential family trend filtering on lattices.

S.9. Additional Simulation Results. This appendix provides additional simulation results complementing Section 6 of the main paper. It includes detailed visual summaries and diagnostics across all DGPs and sample sizes.

S.9.0.0.1. Contents of this appendix.

- Asymptotic normality and variance estimation (Section S.9.1): combined full–page coverage and CI–width panels across DGPs and sample sizes.
- Asymptotic efficiency (Section S.9.2): efficiency comparisons for TN, GS3, GA3, GS5, Step, and Sine.
- EIC–based variance estimation for statistical estimands (Section S.9.3): per–DGP CI width and coverage figures.

S.9.1. *Pointwise Asymptotic normality and variance estimation.* This subsection provides full pointwise asymptotic normality and variance estimation panels for all six DGPs.

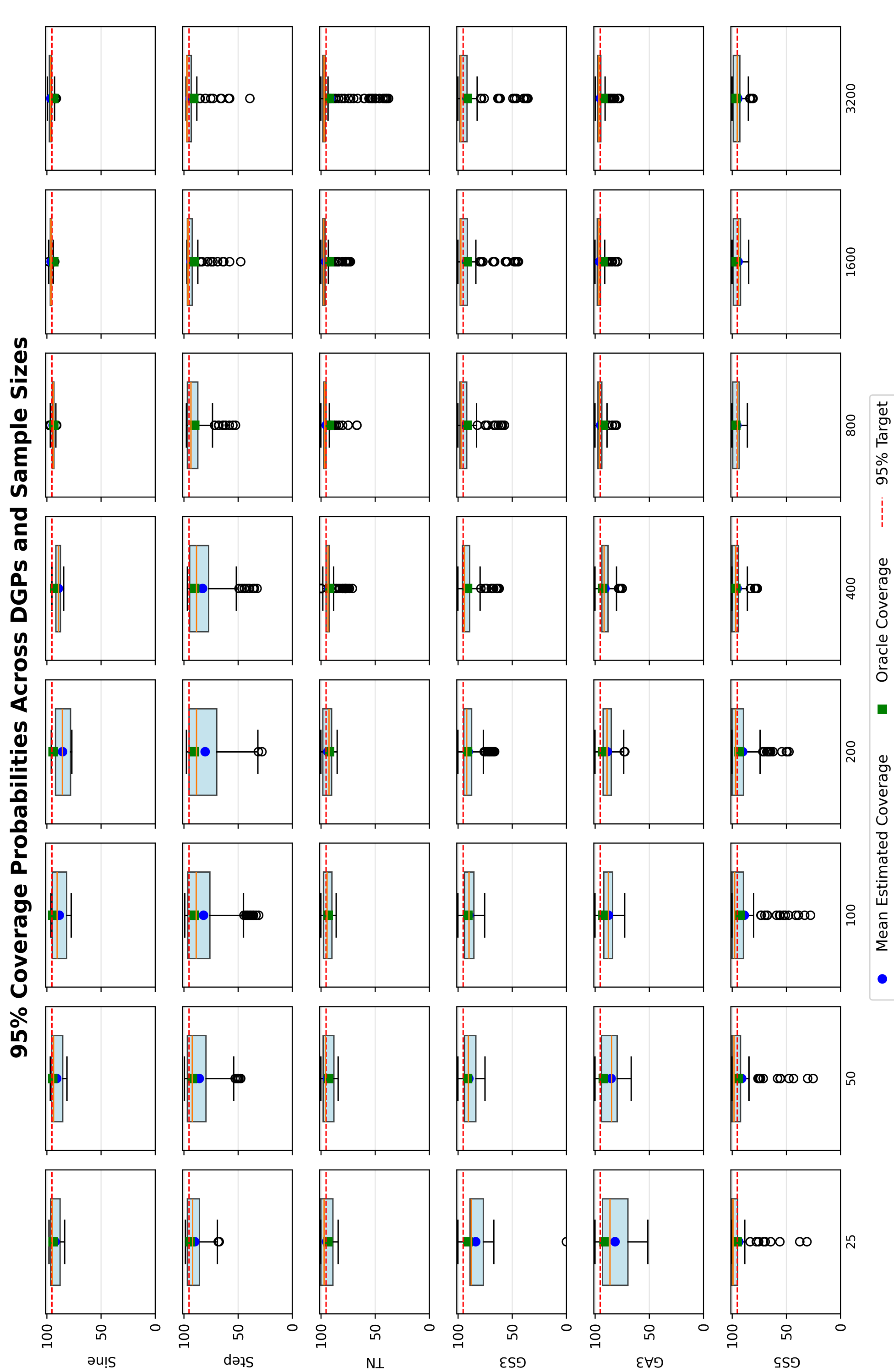

FIG 1. *95% coverage probabilities across DGPs and sample sizes (rotated for full–page display).*

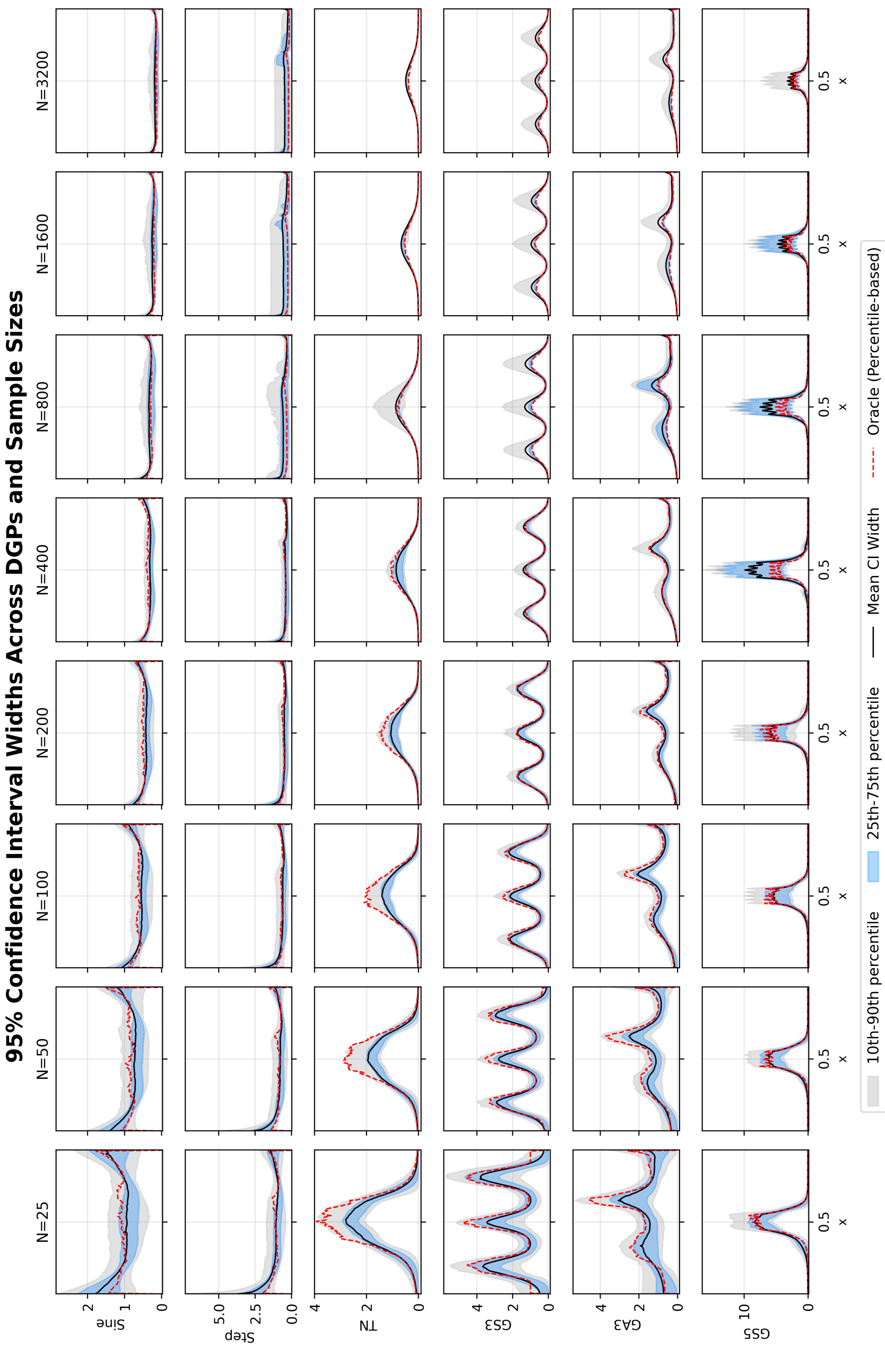

FIG 2. *95% confidence–interval widths across DGPs and sample sizes (rotated for full–page display).*

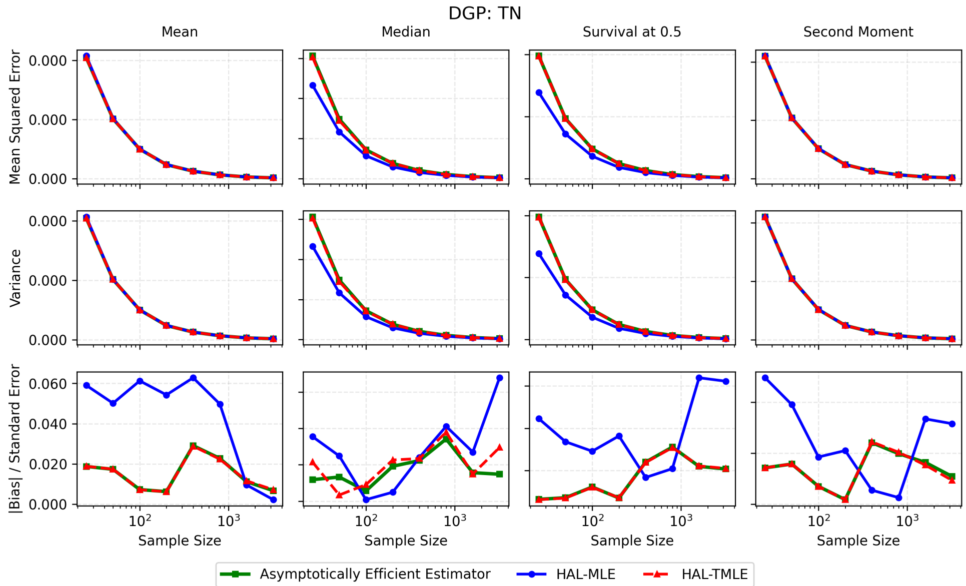

FIG 3. *Asymptotic–efficiency comparison for TN.*

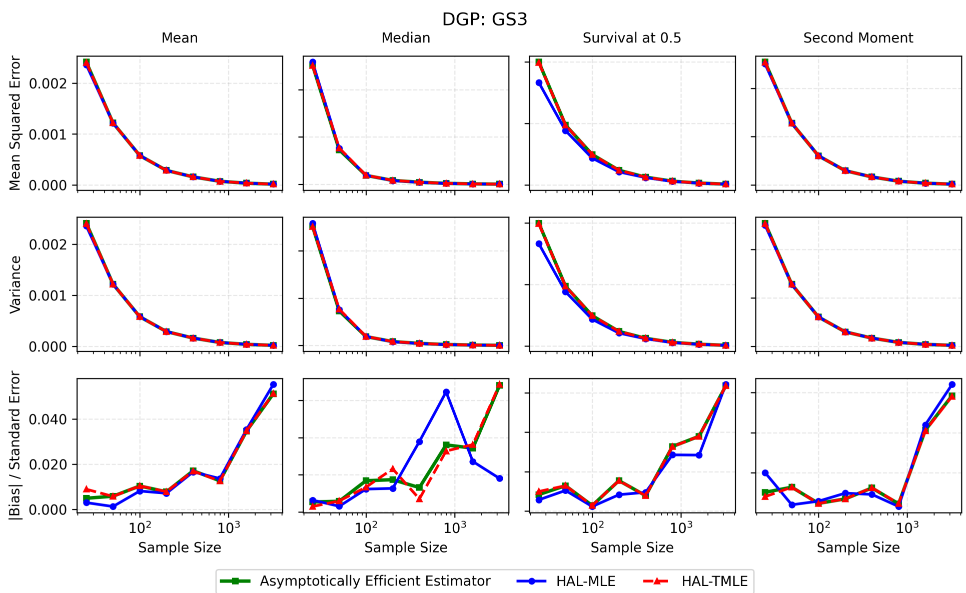

FIG 4. *Asymptotic–efficiency comparison for GS3.*

S.9.2. *Asymptotic efficiency.* This subsection provides full asymptotic-efficiency comparison panels for all six DGPs.

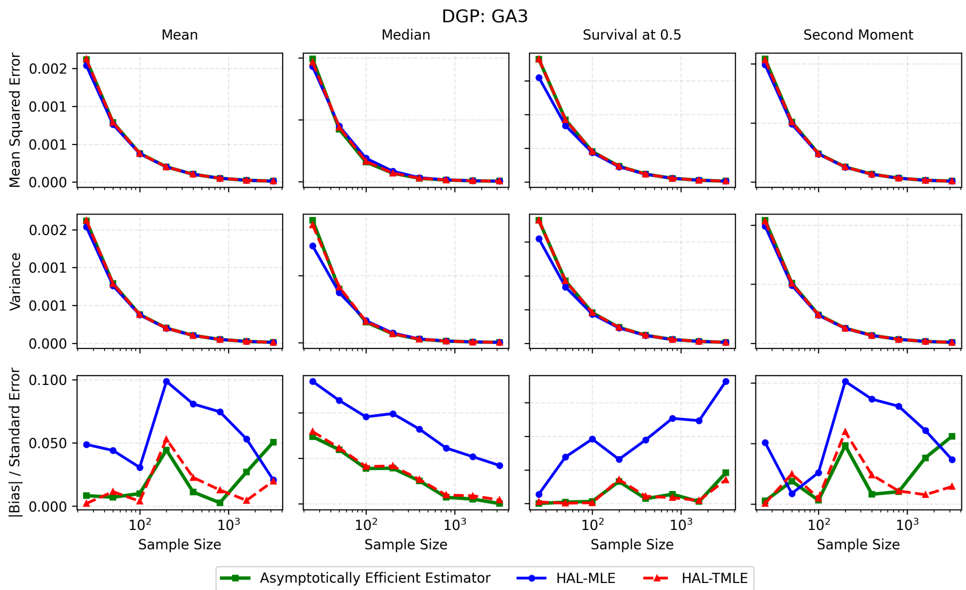

FIG 5. *Asymptotic–efficiency comparison for GA3.*

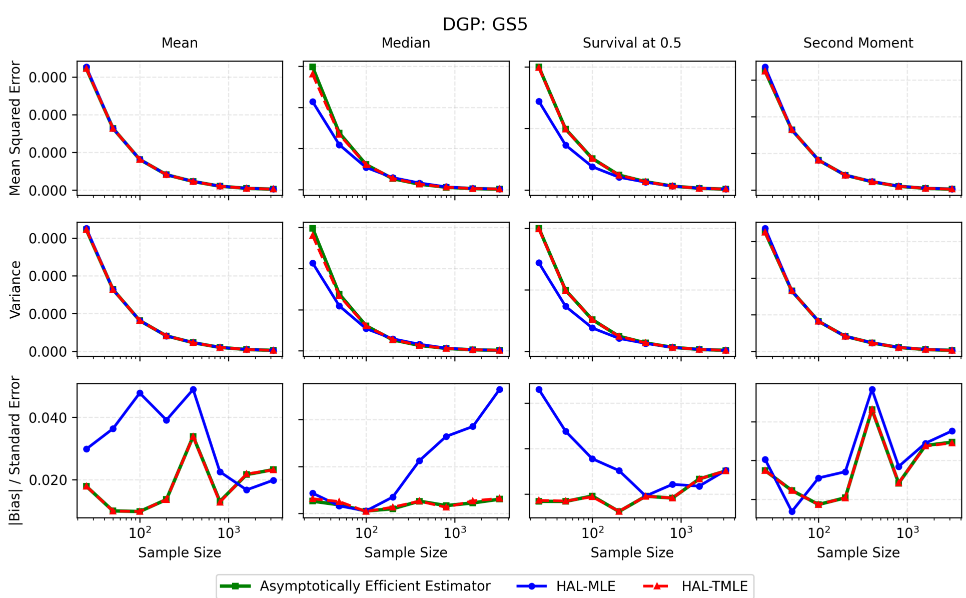

FIG 6. *Asymptotic–efficiency comparison for GS5.*

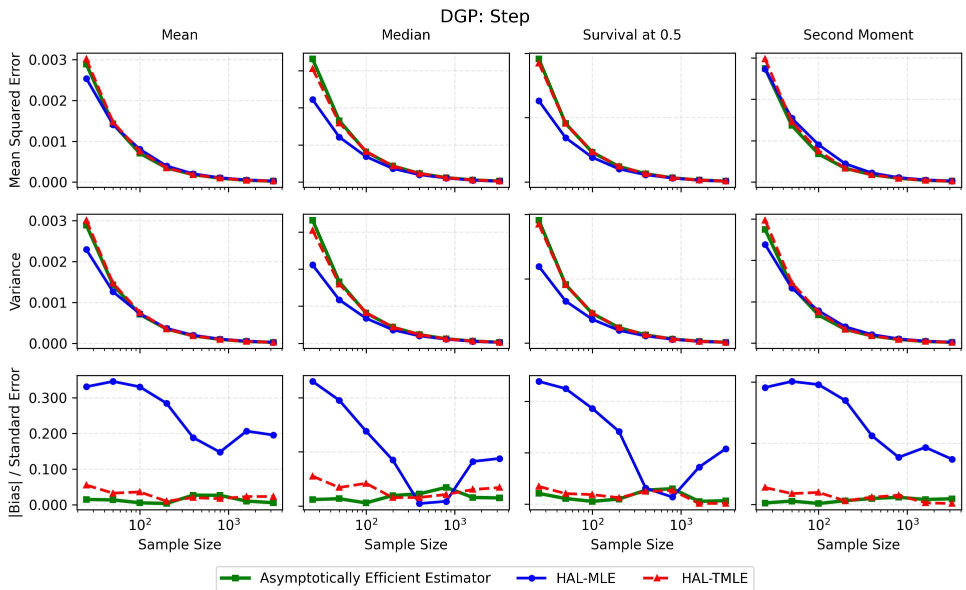

FIG 7. *Asymptotic–efficiency comparison for Step.*

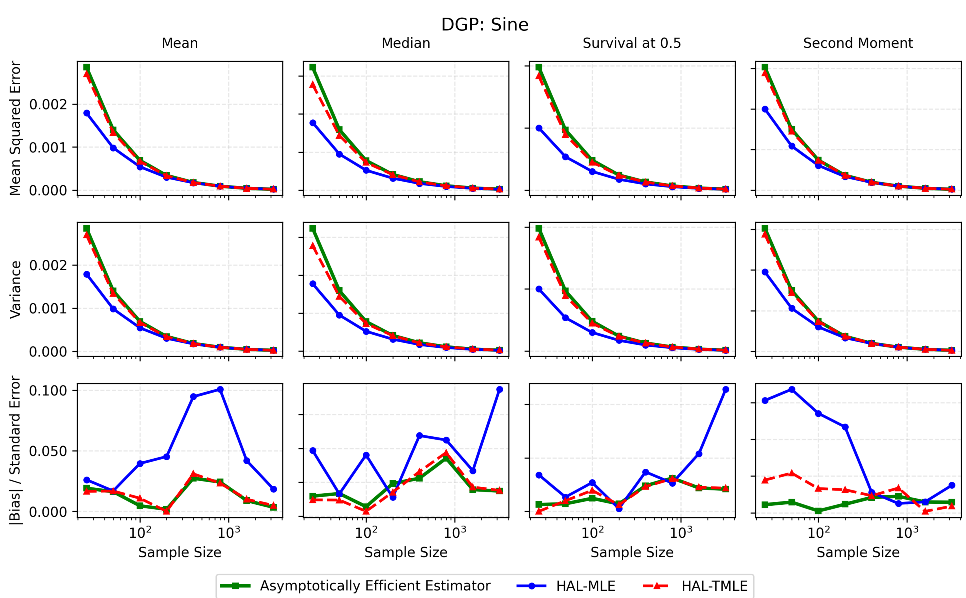

FIG 8. *Asymptotic–efficiency comparison for Sine.*

DGP	N=25	N=50	N=100	N=200	N=400	N=800	N=1600	N=3200
TN	(93.3, 94.6)	(94.4, 96.0)	(94.9, 95.2)	(95.8, 95.4)	(94.7, 94.7)	(93.9, 95.1)	(95.3, 95.2)	(94.3, 94.3)
GS3	(95.1, 95.9)	(95.2, 95.8)	(95.7, 95.9)	(95.5, 94.9)	(94.3, 95.1)	(95.3, 94.9)	(95.1, 94.6)	(96.3, 96.1)
GA3	(94.8, 95.4)	(94.7, 95.3)	(95.6, 94.6)	(95.3, 95.0)	(94.9, 95.2)	(94.3, 94.2)	(96.2, 94.9)	(94.8, 94.6)
GS5	(95.5, 96.0)	(94.3, 94.7)	(96.2, 96.0)	(94.7, 94.5)	(93.1, 93.6)	(94.9, 94.8)	(95.5, 94.4)	(94.6, 94.0)
Step	(93.5, 95.1)	(95.6, 96.4)	(92.6, 93.7)	(95.6, 95.3)	(94.0, 94.6)	(94.0, 95.1)	(95.2, 95.0)	(94.4, 94.3)
Sine	(92.8, 93.6)	(95.1, 95.1)	(94.5, 94.7)	(95.8, 95.4)	(94.7, 94.9)	(94.3, 95.3)	(95.3, 95.3)	(94.4, 94.1)

TABLE 2
Coverage for mean

DGP	N=25	N=50	N=100	N=200	N=400	N=800	N=1600	N=3200
TN	(93.2, 94.7)	(94.8, 95.4)	(94.1, 94.8)	(95.6, 95.1)	(94.5, 94.9)	(93.9, 95.1)	(95.2, 95.2)	(94.5, 94.3)
GS3	(94.9, 95.3)	(94.9, 95.6)	(95.6, 95.7)	(95.4, 95.0)	(94.0, 94.4)	(94.6, 94.8)	(95.2, 94.7)	(96.7, 95.1)
GA3	(94.3, 95.6)	(94.4, 95.1)	(95.8, 95.1)	(95.0, 95.0)	(95.5, 95.8)	(93.9, 94.3)	(95.8, 95.1)	(95.6, 95.1)
GS5	(95.5, 96.0)	(94.2, 94.8)	(96.0, 96.1)	(95.3, 95.1)	(93.2, 94.1)	(94.8, 94.8)	(95.2, 94.4)	(94.9, 94.1)
Step	(92.8, 96.0)	(94.8, 96.9)	(91.4, 94.4)	(94.9, 95.7)	(93.5, 94.0)	(94.0, 95.1)	(95.2, 95.1)	(94.2, 93.8)
Sine	(93.3, 94.2)	(94.6, 95.0)	(94.1, 94.6)	(95.2, 95.4)	(93.4, 93.5)	(94.2, 95.1)	(94.7, 94.7)	(94.7, 94.0)

TABLE 3
Coverage for second moment

DGP	N=25	N=50	N=100	N=200	N=400	N=800	N=1600	N=3200
TN	(95.4, 95.4)	(94.4, 94.4)	(95.0, 95.0)	(95.4, 95.4)	(94.4, 94.7)	(95.4, 95.5)	(95.5, 95.5)	(95.2, 94.6)
GS3	(95.9, 95.9)	(93.6, 93.6)	(94.8, 94.8)	(95.0, 95.0)	(94.5, 94.7)	(95.8, 95.5)	(93.8, 93.8)	(96.5, 94.5)
GA3	(94.2, 96.5)	(96.0, 95.3)	(93.2, 94.8)	(93.9, 94.9)	(95.3, 95.3)	(95.6, 95.0)	(96.1, 95.4)	(95.8, 95.5)
GS5	(96.1, 96.1)	(93.1, 94.3)	(94.4, 95.1)	(95.9, 95.7)	(94.3, 95.1)	(94.8, 95.4)	(94.8, 94.7)	(95.1, 94.8)
Step	(93.4, 96.2)	(95.1, 96.8)	(96.2, 95.8)	(94.8, 95.5)	(93.9, 94.6)	(94.3, 94.5)	(96.0, 96.0)	(95.0, 94.2)
Sine	(95.4, 95.4)	(94.5, 94.5)	(95.1, 95.1)	(95.4, 95.4)	(94.4, 94.4)	(94.9, 95.6)	(95.3, 95.4)	(95.2, 94.6)

TABLE 4
Coverage for survival at 0.5

DGP	N=25	N=50	N=100	N=200	N=400	N=800	N=1600	N=3200
TN	(91.5, 94.3)	(94.0, 94.8)	(95.5, 95.9)	(95.0, 94.9)	(94.7, 95.5)	(94.8, 95.5)	(95.4, 95.6)	(95.4, 94.5)
GS3	(93.8, 92.8)	(97.0, 95.8)	(97.7, 94.7)	(96.6, 95.3)	(94.7, 94.9)	(94.9, 95.2)	(93.6, 94.1)	(96.2, 95.1)
GA3	(89.8, 91.3)	(92.4, 93.8)	(93.7, 94.5)	(96.3, 94.7)	(96.7, 94.4)	(96.1, 94.8)	(96.7, 95.1)	(95.0, 94.8)
GS5	(91.2, 94.8)	(92.7, 93.4)	(95.6, 94.7)	(96.7, 94.9)	(96.5, 94.0)	(96.3, 95.9)	(95.2, 94.8)	(96.4, 94.6)
Step	(91.8, 95.2)	(93.3, 95.5)	(94.7, 96.3)	(93.7, 94.6)	(94.2, 95.4)	(95.1, 95.6)	(95.8, 96.3)	(95.8, 95.2)
Sine	(91.9, 94.0)	(93.7, 94.1)	(95.1, 94.5)	(94.2, 94.5)	(95.2, 95.2)	(94.7, 95.1)	(95.7, 96.0)	(96.1, 95.3)

TABLE 5
Coverage for median

S.9.3. *EIC–based variance estimation for statistical estimands.* This subsection reports detailed EIC-based coverage tables and CI-width/coverage plots for all six DGPs.

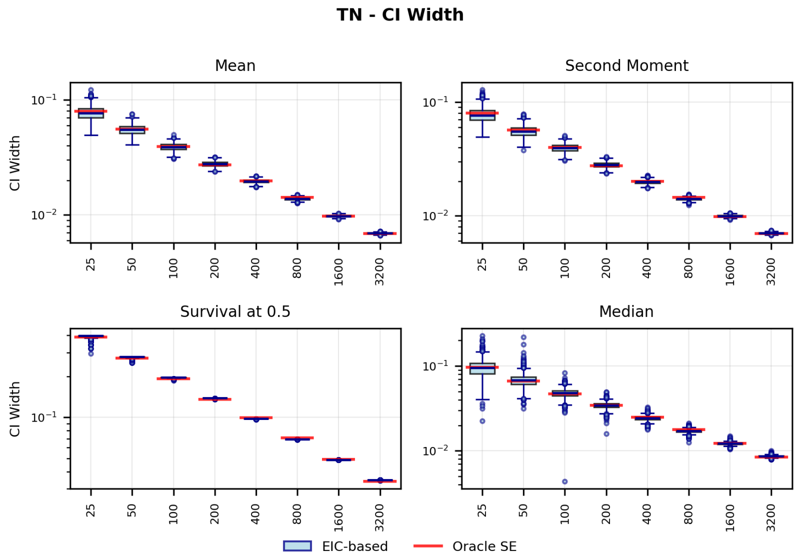

FIG 9. *EIC–based variance estimation for TN: CI width.*

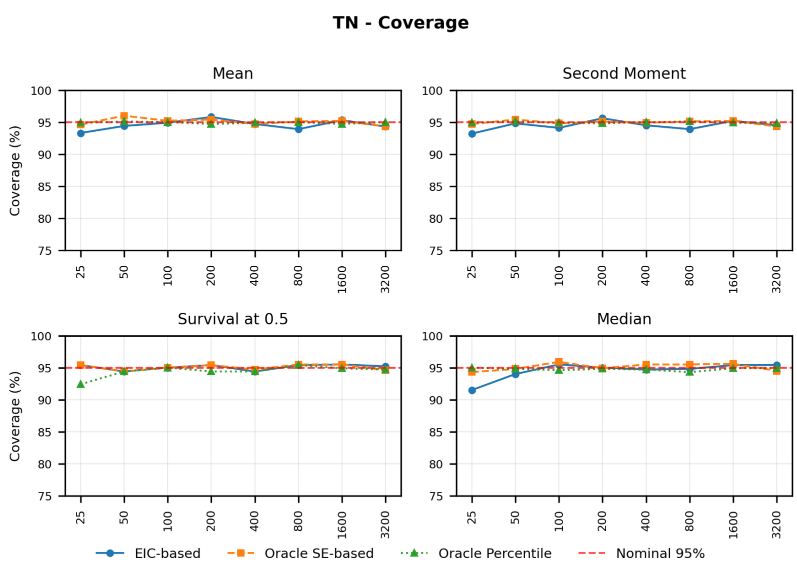

FIG 10. *EIC–based variance estimation for TN: coverage.*

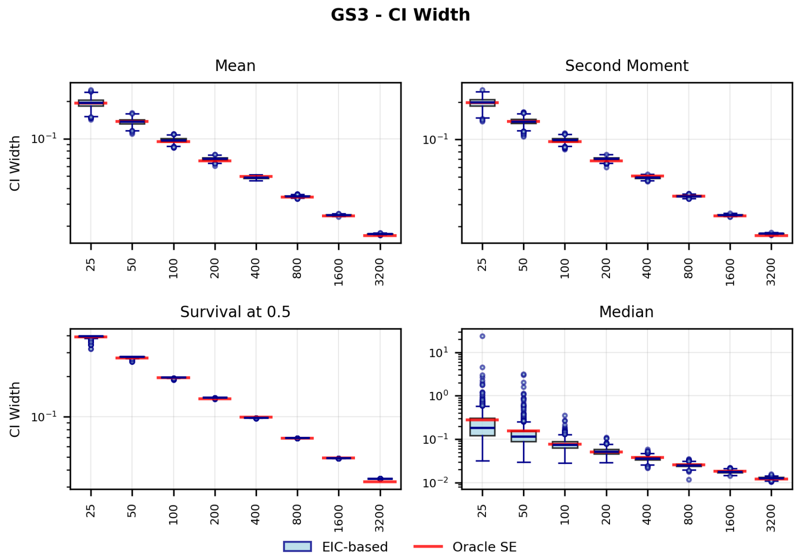

FIG 11. *EIC–based variance estimation for GS3: CI width.*

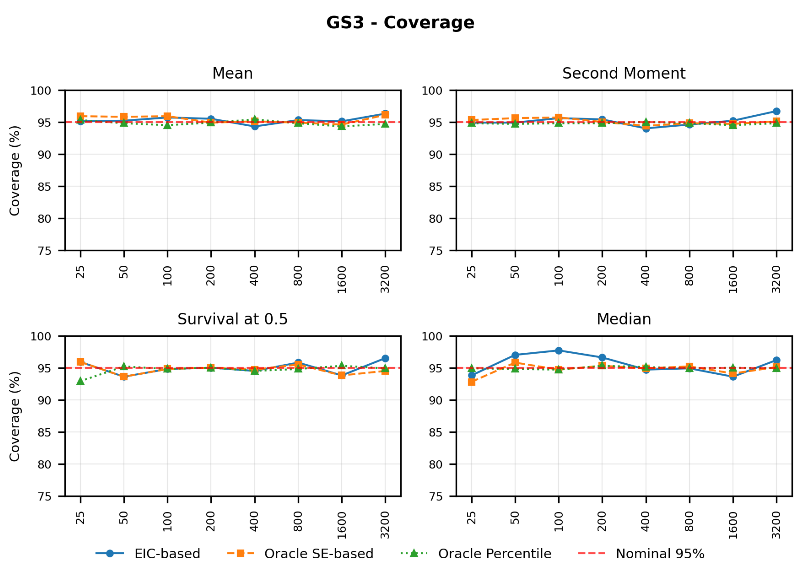

FIG 12. *EIC–based variance estimation for GS3: coverage.*

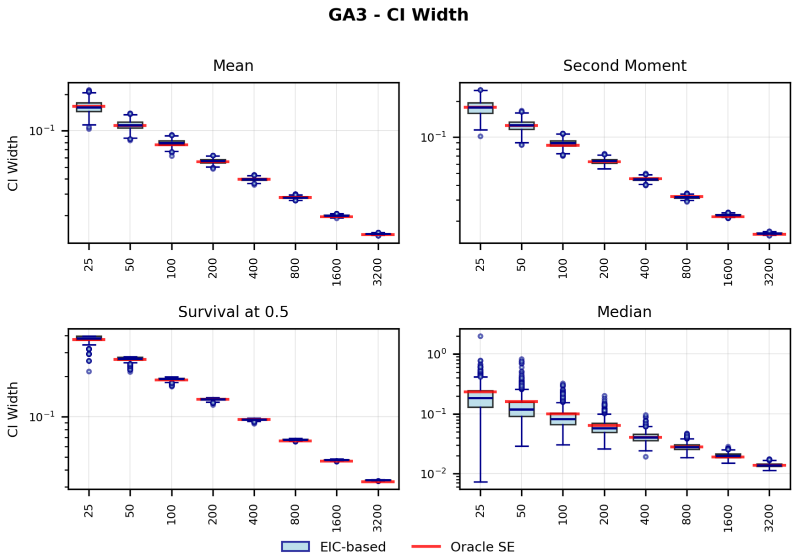

FIG 13. *EIC–based variance estimation for GA3: CI width.*

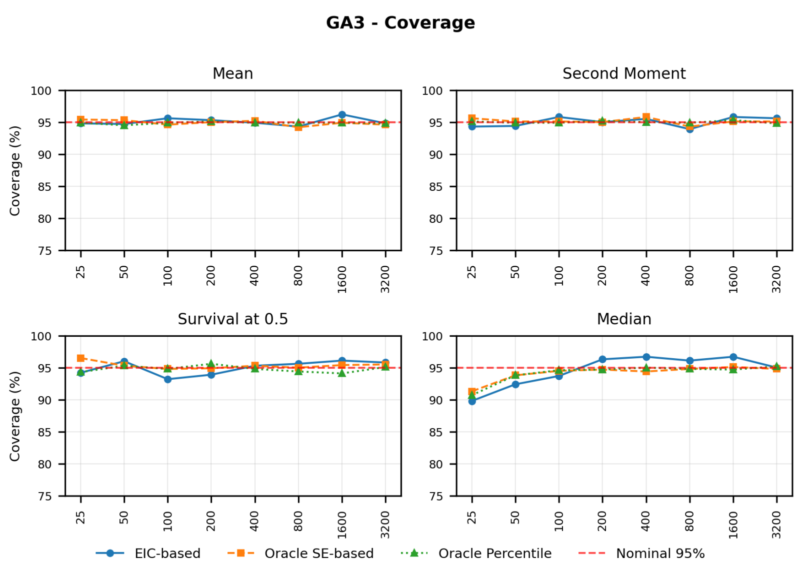

FIG 14. *EIC–based variance estimation for GA3: coverage.*

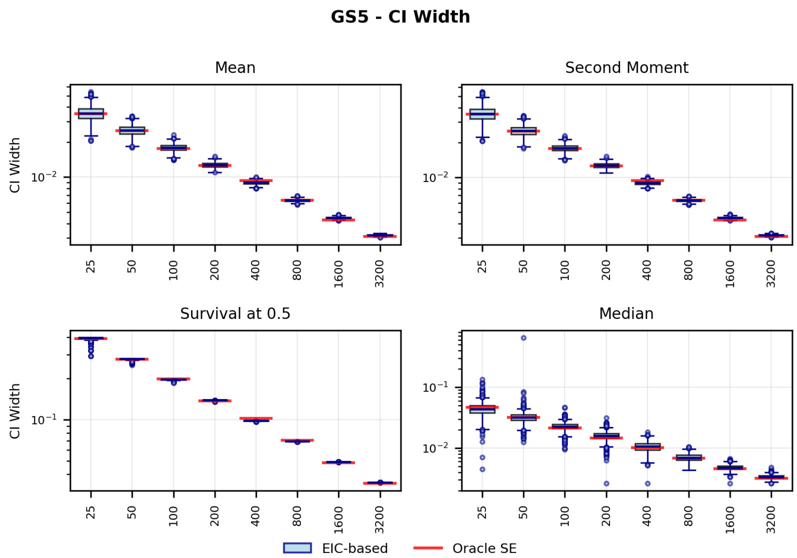

FIG 15. *EIC–based variance estimation for GS5: CI width.*

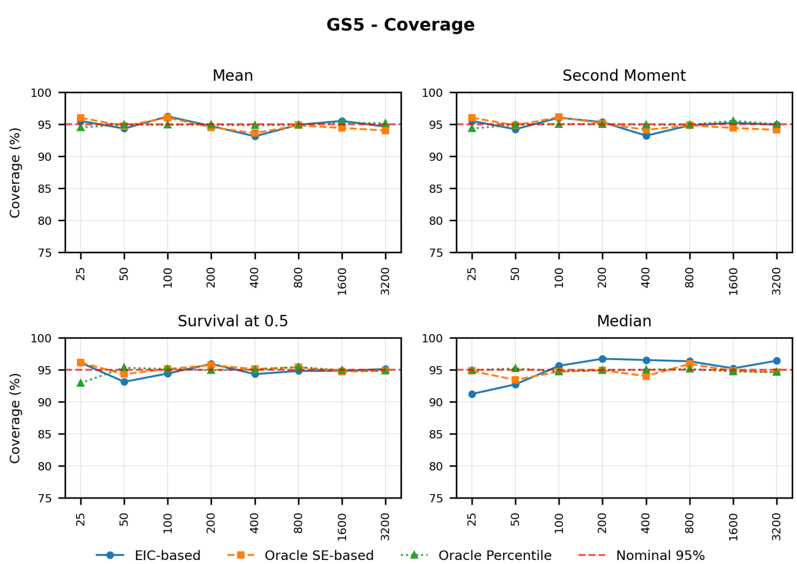

FIG 16. *EIC–based variance estimation for GS5: coverage.*

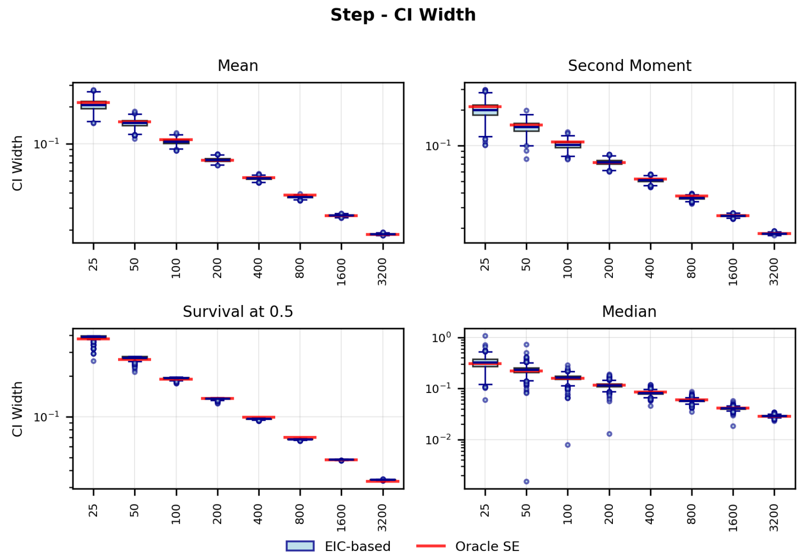

FIG 17. *EIC–based variance estimation for Step: CI width.*

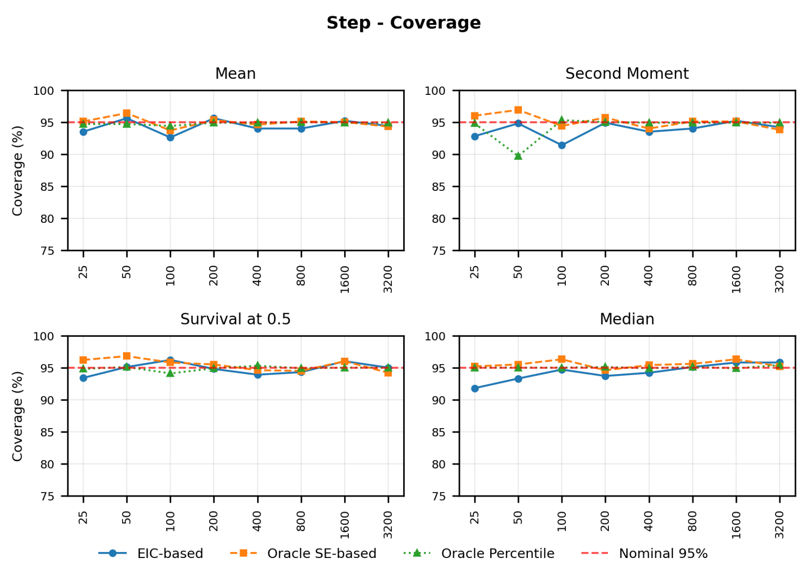

FIG 18. *EIC–based variance estimation for Step: coverage.*

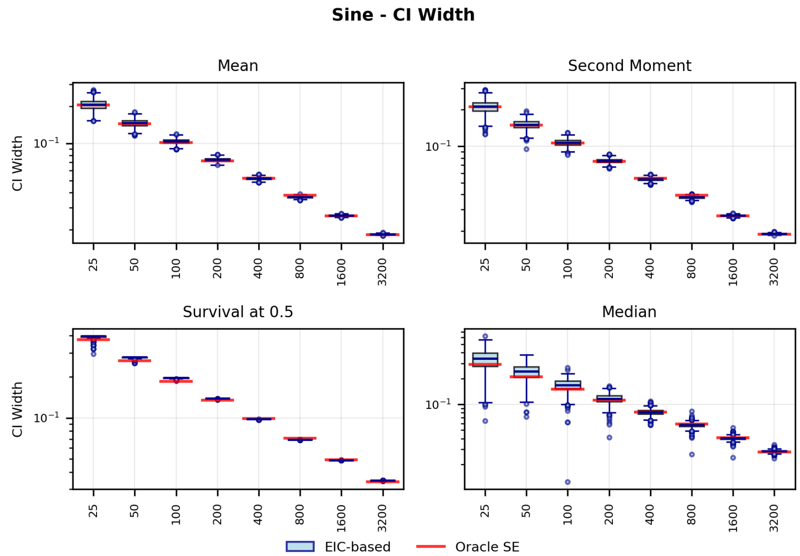

FIG 19. *EIC–based variance estimation for Sine: CI width.*

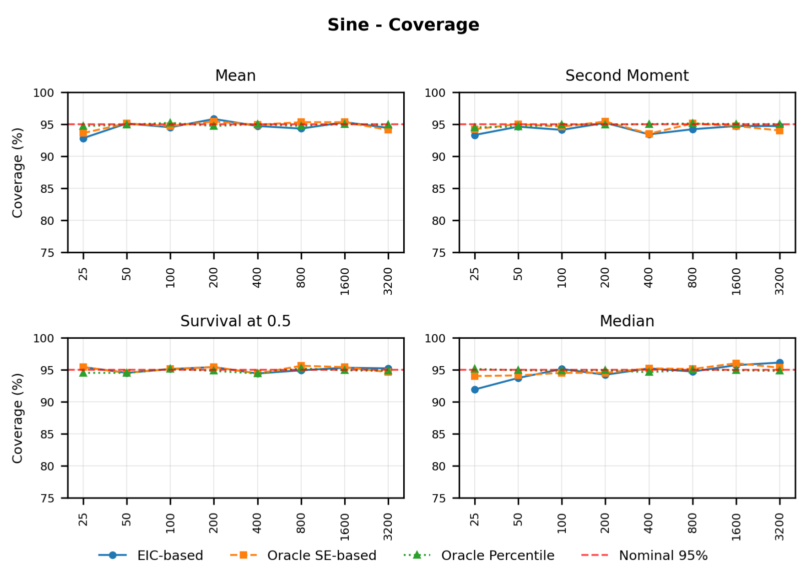

FIG 20. *EIC–based variance estimation for Sine: coverage.*

S.10. Optimization Algorithm Performance Analysis. This appendix provides comprehensive visualization of optimization algorithm performance across all six data-generating processes (DGPs) and basis orders. The figures show both knot selection behavior and loss convergence patterns for four optimization algorithms: FISTA, ProximalAdaGrad, ProximalNewton, and ProximalNewtonLBFGS, compared against the `CVXPY` reference solution.

S.10.1. *Optimization Algorithm.* Recall that we are optimizing the problem with the following form:

$$\beta_n(M) = \arg\max_{\beta:\ \|\beta\|_1 \leq M} \sum_{i=1}^{n} \sum_{j=1}^{n+k+1} \beta_j \phi_j(x_i) - n \log C(\beta), \tag{S.10.1}$$

and its Lagrangian form is as follows:

$$\beta_n(M) = \arg\max_{\beta} \sum_{i=1}^{n} \sum_{j=1}^{n+k+1} \beta_j \phi_j(x_i) - n \log C(\beta) - \lambda \|\beta\|_1, \tag{S.10.2}$$

Tuning the upper bound M of the L_1 norm in the constraint form is more statistically meaningful, because the final choice of M should reflect the magnitude of the SVN of the truth. However, we prefer the Lagrangian form because the algorithm for the constraint form always involves projection to the L_1 ball. In practice, we find that soft thresholding provides a more robust dimension reduction through iterations than projection to the L_1 ball.

S.10.1.1. *Candidate Algorithms.* While generic convex solvers such as `MOSEK`, `ECOS`, or `SCS` within the `CVXPY` package can, in principle, handle our convex optimization problem, their general strategy is to map any instance into a conic programming framework. This approach, though flexible, is not tailored to the structural features of our estimator. Moreover, as the sample size increases to the range of 3,000–5,000 observations, we find that the default knot point selection of these solvers becomes less effective: a large number of unnecessary knot points remain active, primarily due to accuracy constraints in the optimization process. A demonstration of this is shown in Figure 21. These limitations motivate the development of specialized algorithms that exploit the structure of the problem and incorporate effective knot point selection strategies, yielding both improved scalability and enhanced statistical efficiency.

We refer to the matrix of initiated basis evaluated at the data points as the basis matrix. The basis matrix is an $n \times (n + k + 1)$ matrix.

S.10.1.1.1. Standard Algorithm. When using the truncated power basis, the Hessian constructed from the basis matrix is usually ill-conditioned and thus first-order methods like Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) would converge slowly in iterations. Therefore, we propose a standard algorithm combining proximal Newton's method and the coordinate descent algorithm. An advantage of proximal Newton is that we can calculate the Hessian in closed form. Also, we could warm start the algorithm when using cross-validation to choose λ in order to keep Newton's method within the quadratic convergence range.

S.10.1.1.2. Large-scale Adjustment. The proximal Newton's method fails for large-scale problems due to memory issues. When the sample size n is too large, such as 10,000 or 100,000, the initialized Hessian will be $O(n^2)$ and too large to be memorized. For this problem, we propose the Quasi-Newton method (L-BFGS) and coordinate descent. An easy implementation is just memorizing the most recent pair of differences in parameters and their corresponding gradient. Thus, the approximation of the Hessian is a scaled identity matrix, and we update

Algorithm	Dominant per-iteration complexity
FISTA	$\Theta\big((n+G)\,K\big)$
Proximal AdaGrad	$\Theta\big((n+G)\,K\big)$
Proximal Newton L - BFGS	$\Theta\big((1+L_k)\,(n+G)\,K\big)$
Proximal Newton (full)	$\Theta\big(G\,K^2+s\,K^2\big)+\Theta\big((1+L_k)\,(n+G)\,K\big)$

TABLE 6

Dominant per - iteration computational complexity. Here n is the number of samples, G the number of midpoint intervals used for normalization, K the number of basis coefficients, s the active set size in coordinate descent, and L_k the number of line - search objective evaluations at iteration k.

the scale at each iteration. The memory reduces from $O(n^2)$ to $O(n)$. Pragmatically, we find out that including the coordinate descent is more robust for large-scale problems.

Another algorithm we propose is a variant of AdaGrad using the inner product defined in Eq. (4.2) of the main paper. We just compute the diagonal entries of the Hessian for each iteration. Thus, the cost per iteration becomes very cheap, and the memory is just $O(n)$ and totally affordable.

REMARK S.10.1. The development of the large-scale methods seems to be unnecessary in the univariate case, because, in HAL theory like Appendix S.2 and Appendix S.5, we only need about $O^+(n^{\frac{1}{2k+3}})$ knots, and we could choose them based on the percentile of the original sample while remaining the performance. However, in the multivariate case, we need such algorithms because (1) the definition of multivariate percentile could be meaningless and (2) even a small sample size could initialize a large number of parameters $n(2^d-1)$ for d-dimensional data.

S.10.1.2. *Exploration of Optimization Algorithms.* We study the performance of the optimization algorithms in some fixed settings where `CVXPY` solvers have trouble selecting knot points. We here demonstrate the performance of the optimization algorithms on the Truncated Normal Data Generating Process (DGP), and the visualizations are shown in Figure 21. We use the final result of `CVXPY` solutions as the baseline, and we evaluate the performance of the optimization algorithms by comparing the knot selection and loss convergence per iteration and per FLOP. The FLOPs are estimated by the computational complexity of the algorithms, which is shown in Table 6. Comprehensive results for all six DGPs and the computational-complexity details based on our implementation are provided in this appendix.

Based on the results, we find that the Proximal Newton algorithm takes the fewest iterations to converge, and it is also efficient in selecting knot points, which is expected because it is a second-order method and we have the closed-form Hessian. The Proximal Newton L-BFGS algorithm is also efficient in selecting knot points, and it is more efficient than the Proximal Newton algorithm in terms of FLOPs. However, the Proximal Newton L-BFGS algorithm is sometimes unstable in the process of knot selection. AdaGrad is very fast and efficient, but in practice, we find that it has oscillatory behavior as well. FISTA is easy to implement and robust, but it is slow as expected in terms of iterations and wall-clock time.

Generally speaking, comparing to the `CVXPY` solvers, the optimization algorithms we proposed succeed in selecting knot points and converging to the global optimum. This aligns with the HAL-MLE theory that we only need $O^+(n^{\frac{1}{2k+3}})$ knots to achieve the desired performance, and this also informs us that the numerical problem of `CVXPY` solvers is severe when the sample size is large. However, we cannot make a fair comparison between the optimization algorithms and the `CVXPY` solvers in terms of wall-clock time because some of the `CVXPY` solvers are closed-source and some are coded in C, whereas we coded our own solvers in Python.

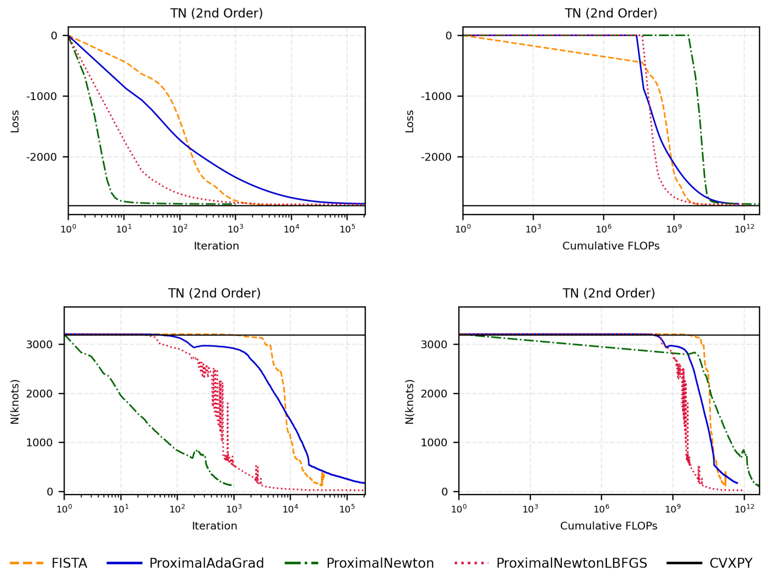

FIG 21. *Optimization algorithm comparison for Truncated Normal (2nd order basis): knot selection per iteration (top left), knot selection per FLOP (top right), loss convergence per iteration (bottom left), and loss convergence per FLOP (bottom right).*

REMARK S.10.2 (Computational comparison with B-splines). It is well known that locally supported B-splines can be more computationally efficient, with their primary advantage stemming from the banded structure of the Hessian matrix. However, this benefit diminishes a lot in large-scale problems where the full Hessian cannot be computed explicitly and must instead be approximated. In addition, for the large-scale problem, the jumping matrix associated with the variation penalty is difficult to track when knot points are chosen in a data-adaptive manner, especially with smoothness order $k \geq 1$. By contrast, the truncated power basis—though less efficient computationally—offers the simplest and most transparent representation of sectional variation, making it particularly well suited for our theoretical analysis and algorithmic development.

S.10.2. *Notation and setup.* We denote by n the number of samples, G the number of midpoint intervals used for normalization on $[0,1]$, $k \in \{0,1,2\}$ the basis order, K the total number of basis coefficients (intercept, k polynomial terms, and M truncated - power terms), s the active set size used in coordinate descent, and L_k the number of objective evaluations performed by backtracking at iteration k. Throughout, matrix - vector multiplies with dense $a \times b$ arrays are costed as $2ab$ FLOPs.

S.10.3. *Per-iteration FLOP derivations.* We summarize implementation - aware FLOP counts for each algorithm; see the project note `experiments/compare_knot_selection/algorithm_flop_per_iter.md` in the experiment git repo for additional commentary and logging details.

S.10.3.0.1. FISTA. A forward pass on data and midpoints and a backward pass via autograd each cost $\Theta\big((n+G)K\big)$, with an additional $\Theta(GK)$ for the exact normalization update. Dominant term: $\Theta\big((n+G)K\big)$.

S.10.3.0.2. Proximal AdaGrad. Computing the gradient uses $\Theta\big(nK+GK\big)$; forming the diagonal Hessian via variance on midpoints contributes another $\Theta(GK)$ (tighter bound) to $\Theta(3GK)$ (conservative). Including the normalization step yields a dominant $\Theta\big((n+G)K\big)$.

S.10.3.0.3. Proximal Newton (full). Each iteration forms the gradient in $\Theta\big((n+G)K\big)$, the Hessian as a weighted covariance on midpoints in $\Theta\big(GK^2\big)$, solves the proximal subproblem via coordinate descent in $\Theta\big(sK^2\big)$, and performs L_k objective evaluations contributing $\Theta\big(L_k(n+G)K\big)$; normalization adds $\Theta(GK)$. Dominant term: $\Theta\big(GK^2+sK^2\big)+\Theta\big((1+L_k)(n+G)K\big)$.

S.10.3.0.4. Proximal Newton L - BFGS. Replacing the full Hessian with an L - BFGS update avoids the GK^2 term; per iteration is dominated by the gradient and line search plus normalization: $\Theta\big((1+L_k)(n+G)K\big)$.

S.10.4. *From iterations to cumulative FLOPs.* Given per - iteration FLOP estimates $F(k)$, cumulative FLOPs up to iteration t are $C(t)=\sum_{k=1}^{t} F(k)$. Our visualization utilities infer L_k from logged step sizes α_k (with backtracking factor $\beta=0.5$) and use consistent baselines at iteration/FLOP zero. See the note for practical defaults and alignment details.

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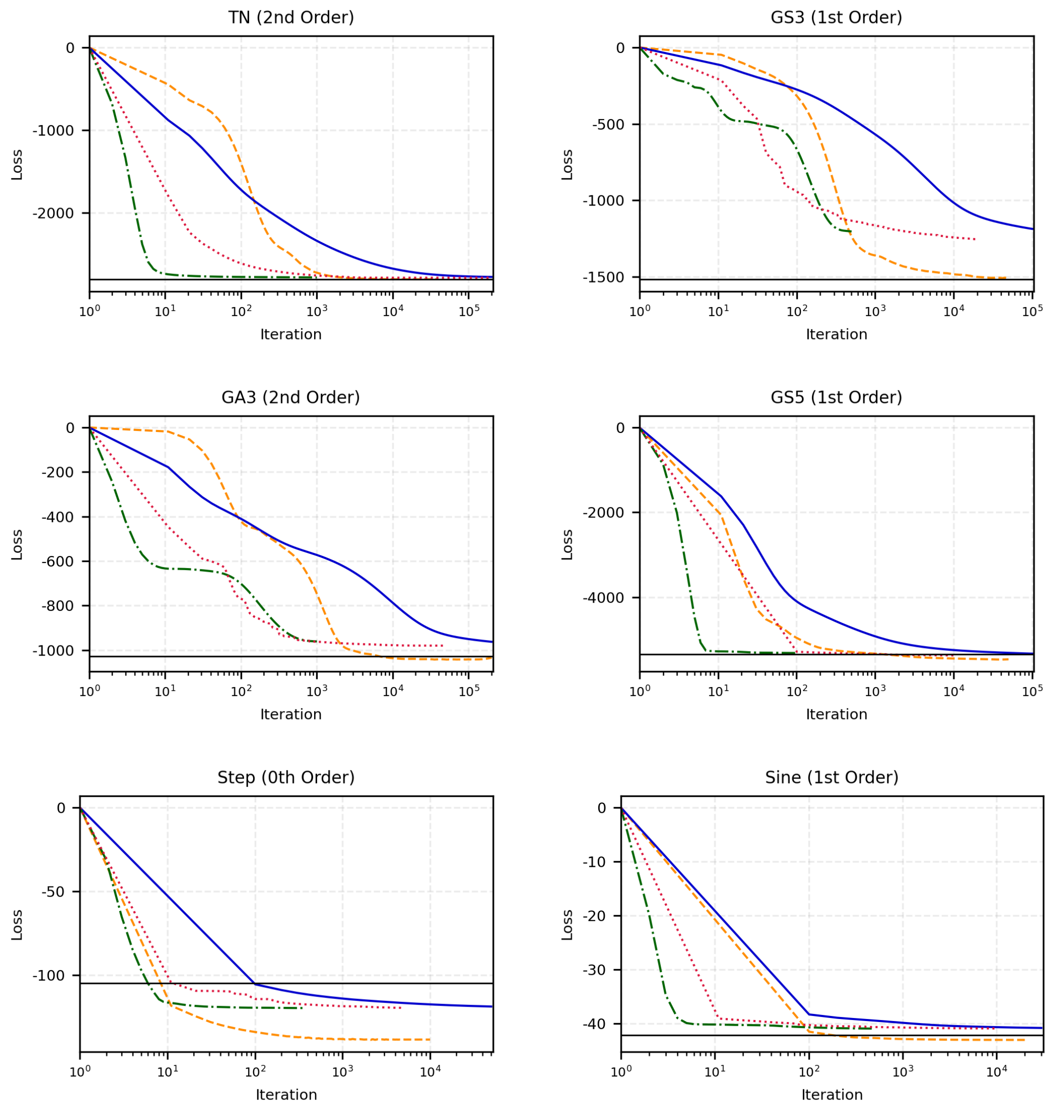

FIG 22. *Per-iteration loss convergence for optimization algorithms across six DGPs. Panels (row-wise, left to right): (a) TN, (b) GS3, (c) GA3, (d) GS5, (e) Step, (f) Sine.*

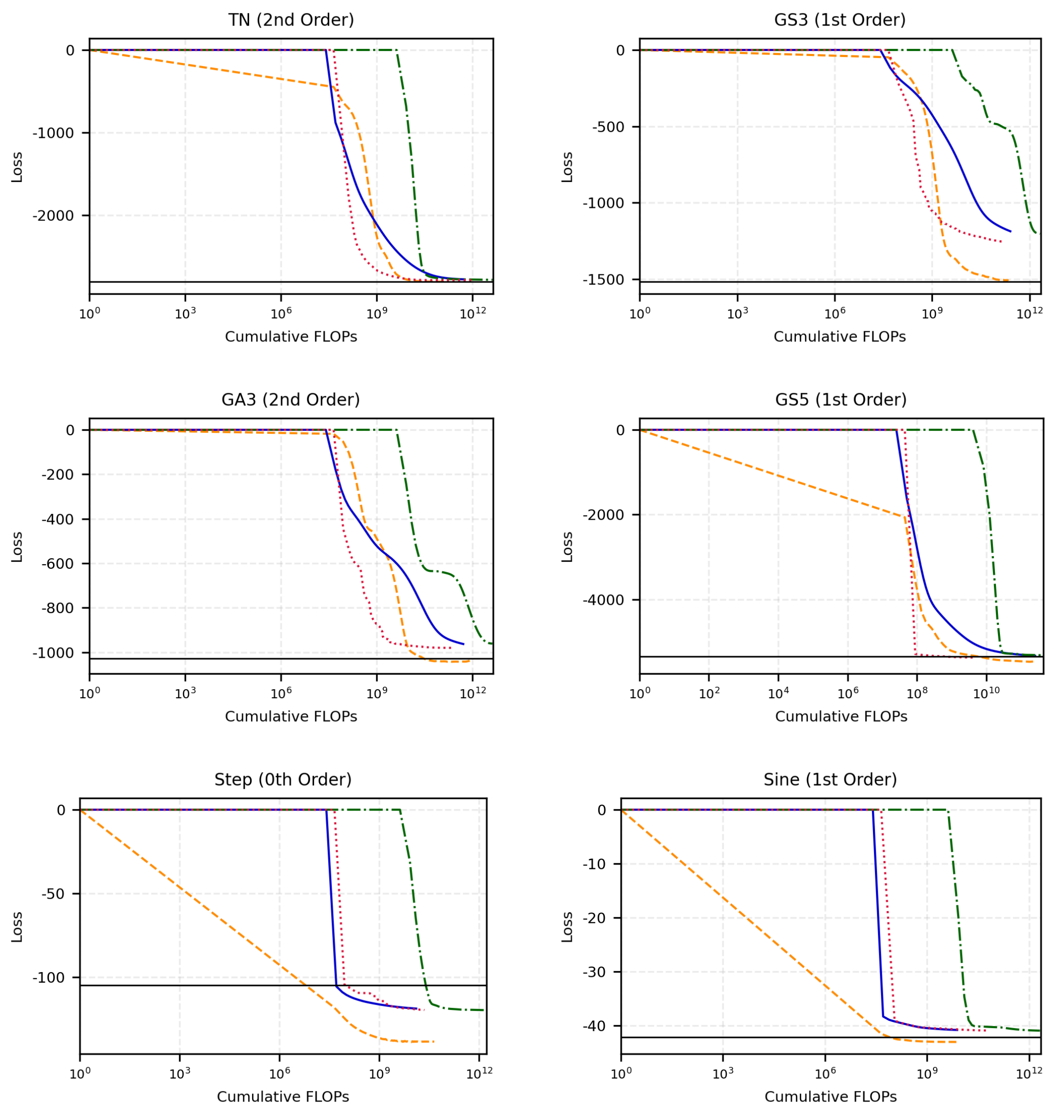

FIG 23. *Per-FLOP-normalized loss convergence across six DGPs. Panels (row-wise, left to right): (a) TN, (b) GS3, (c) GA3, (d) GS5, (e) Step, (f) Sine.*

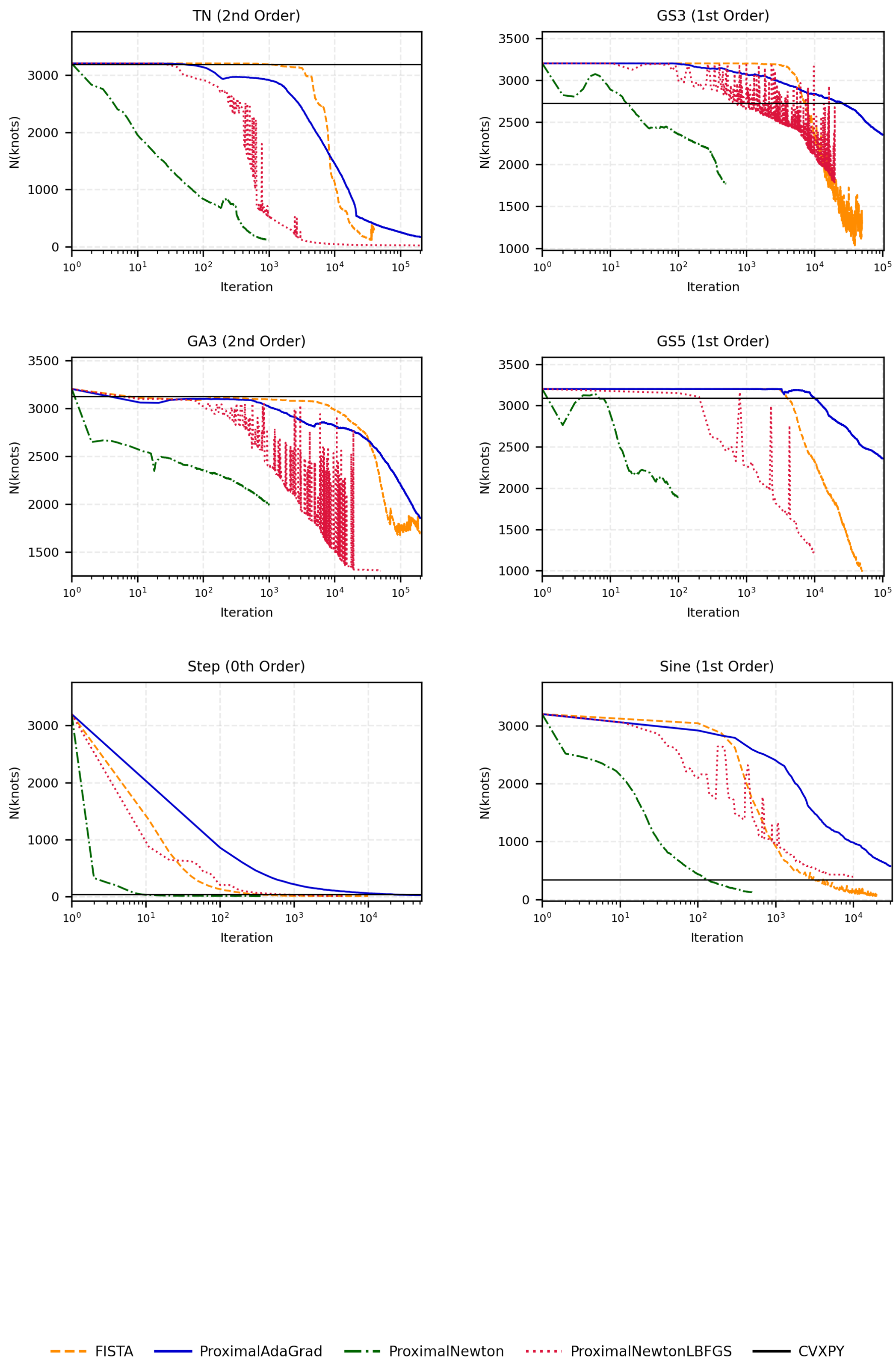

FIG 24. *Per-iteration knot selection for optimization algorithms across six DGPs. Panels (row-wise, left to right): (a) TN, (b) GS3, (c) GA3, (d) GS5, (e) Step, (f) Sine.*

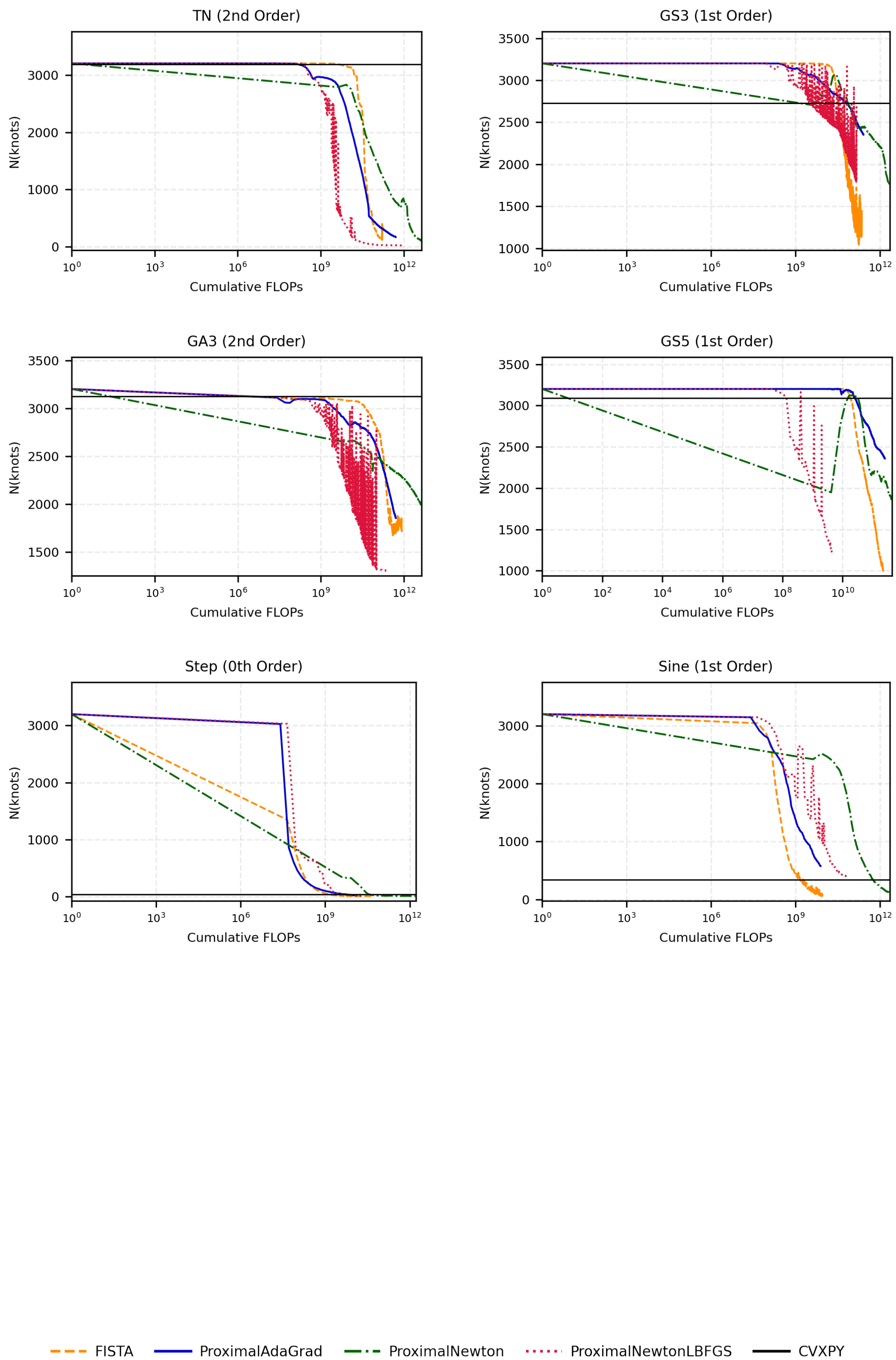

FIG 25. *Per-FLOP-normalized knot selection across six DGPs. Panels (row-wise, left to right): (a) TN, (b) GS3, (c) GA3, (d) GS5, (e) Step, (f) Sine.*